

Amalgam Restorations

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The word *amalgam* means an alloy of mercury with another metal or metals.[1] This type of alloying is called *amalgamation*.[2] In dentistry, before these metals are combined with mercury to make dental amalgam, they are known as *dental amalgam alloys*. Prior to the development of high-copper amalgam alloys, dental amalgam alloys contained at least 65% by weight silver, 29% by weight tin, and less than 6% by weight copper. The high-copper amalgam alloys[3,4] contain between 12% and 30% by weight copper and at least 40% by weight silver.[5] This higher level of copper has resulted in the elimination of the highly corrodible and weak gamma 2 (tin-mercury) phase that existed in the low-copper dental amalgams.[6–8]

Zinc is added to amalgam to enhance its mechanical properties,[9] reduce marginal fracture,[10,11] and prolong the service of the restoration.[11,12] When moisture is incorporated during condensation of a zinc-containing low-copper amalgam, delayed expansion will occur.[13] Zinc-containing high-copper amalgams do not exhibit the phenomenon of delayed expansion,[14–16] but isolation to prevent any moisture contamination is important for both zinc-containing and zinc-free amalgam

restorations.[14] Contamination of dental amalgam with moisture will create porosity in the restoration, which will decrease strength and increase both corrosion and creep.[14,16]

This process by which amalgam alloy powder is mixed with liquid mercury is called *trituration*.[17] The powder may be of the lathe-cut variety (Fig 12-1a), which is made by milling an ingot of the alloy, or of the spherical type (Fig 12-1b), which is made by atomizing liquid alloy.[18] The spherical particles usually are not true spheres but take on various rounded shapes. Some dental amalgam alloys contain only lathe-cut particles, called *filings*; others contain only spheres, and some contain both spheres and filings. Those containing only filings are called *lathe-cut amalgam alloys*, those containing only spheres are called *spherical alloys*, and those containing both filings and spheres are called *admixtures*[5] (Fig 12-1c).

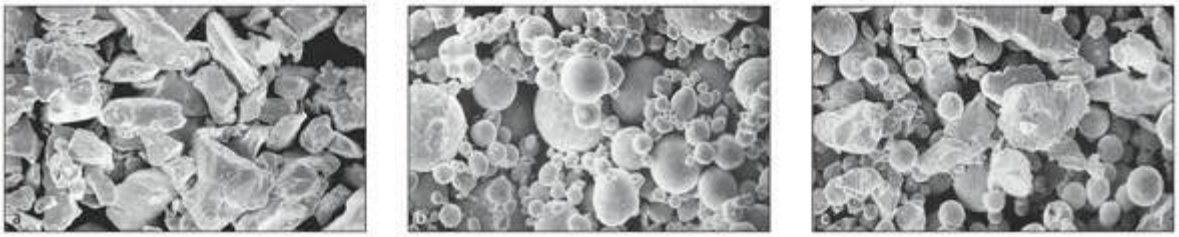


Fig 12-1 *(a)* Lathe-cut amalgam alloy particles. *(b)* Spherical amalgam alloy particles. *(c)* Admixed amalgam alloy particles.

Dental amalgam restorations contain approximately 50% mercury, and some people have concerns that mercury, which is known to be toxic when present in certain forms and in high doses, is present in the mouth. Mercury in amalgam is bound in the matrix phase. The metallic bonds in amalgam are very difficult to break, and only very heavy pressure or high heat can potentially cause the bonds in the restoration to degrade. During mastication, pressures of 200 MPa or higher are common, and this force, with friction, can generate heat in small areas. It has been theorized[19] that this heat could cause amalgam to release small amounts of mercury during function.[20] It has been estimated that for a patient having 12 amalgam restorations in his or her mouth, the amount of metallic mercury released per day would only be in the range of 1.7 µg.[21]

There is considerable evidence of the safety of dental amalgam.[19,22–27] To date, there is no confirmed evidence to indicate that the mercury in dental amalgam is related to any disease.[28–34] Research reveals that no toxic effect has been linked to the level of mercury released from amalgam, even when amalgam restorations are removed.[35,36] Several agencies, including the World Health Organization,[37] the US

Public Health Service,[38] the National Institutes of Health,[39] and the Swedish Medical Research Council,[40] have reaffirmed their positions that there are no data to compel a change in the current use of dental amalgam.[41,42] While no evidence refutes these claims, a general consensus exists to reduce mercury use in all industries because of concerns over environmental effects.[43]

Any component of amalgam or any other restorative material can elicit an allergic reaction, but hypersensitivity to mercury is extremely rare. Of those who have an established allergy to mercury, fewer than 1% demonstrate clinically observable reactions to mercury in dental amalgam restorations.[44] Raap et al[45] examined 206 patients thought to have a contact allergy to dental metals. Only 28 of the 206 patients examined had positive patch test reactions to metals used in dentistry, and some of these had reactions to more than one of the tested metals. Ten were positive to gold sodium thiosulfate, ten to palladium chloride, ten to nickel sulfate, and four to amalgam. In case reports on amalgam hypersensitivity, the presentation has been oral lichenoid lesions, erythematous lesions, burning, and/or itching.[46]

Dental amalgam has served as a dental restorative for more than 175 years; it was used as early as 1820 in Europe, and by the mid-1830s, it was in use in the United States.[42,44] In a survey conducted in 2005,[47] more than two-thirds of North American dentists who responded stated that they used dental amalgam. While the percentage is undoubtedly lower today, dental amalgam continues to be used extensively.

Advantages and Disadvantages

Dental amalgam has many advantages as a restorative material. It is strong, durable, and relatively easy to use. It wears at a rate similar to that of tooth structure. Over time, amalgam has the ability to bring about, through formation of corrosion products, reduced microleakage at its interface with tooth structure.[6,48] Resin composite placed at and below the gingival crest has been found to alter the subgingival microflora in favor of a more pathogenic mix.[49] Subgingival amalgam margins did not significantly alter the bacterial biofilm under the same conditions.[49] A review of insurance claim data indicated that molars restored with amalgam restorations required endodontic therapy much less frequently than those restored with resin composite50; however, this review was done prior to the development and implementation of current bonding systems, so future reviews may present different findings. When replacing restorations, it is much easier to avoid enlarging the cavity preparation during removal of amalgam than it is when removing resin

composite because of the contrast between the color of tooth structure and amalgam and because amalgam is not typically bonded to the walls of the preparation.[51–53] In contrast, removal of resin composite materials consistently results in a larger preparation than the original.[54] Of the long-term restorative materials, dental amalgam is the least time-consuming to place and has the lowest cost.

Clinical research does not support the often-made claim that dental amalgam causes cuspal fracture. Wahl et al[55] have presented data showing that teeth restored with amalgam do not fracture any more frequently than teeth restored with resin composite. Weakening of tooth structure by large restorative tooth preparations is a likelier cause of fracture in tooth structure adjacent to restorations.

Dental amalgam also has disadvantages. Its primary disadvantage is that it is not tooth-colored. In addition, it does not, on its own, bond to tooth structure, although amalgam bonding systems have been proven to provide a fairly reliable mechanical attachment of amalgam to enamel and dentin.[56] Amalgam must be handled properly to avoid excessive exposure of the dental staff to mercury vapor.[57,58] Despite its disadvantages, however, dental amalgam is an economic and reliable direct restorative material that is still used in a majority of dental practices. When placed properly in well-designed tooth preparations, it will serve well for long periods of time.[12,59,60] Many studies have demonstrated the excellent longevity of dental amalgam.[11,59,61–63]

This chapter addresses Class 1 and Class 2 preparations and placement techniques for dental amalgam. Complex restorations involving the replacement of cusps with dental amalgam are also described. Class 5 amalgam preparations and restorations are covered in chapter 15.

Resistance Form

There are two considerations for resistance form when a tooth is being prepared to receive an amalgam restoration. First, resistance form should be developed for the restoration; the restoration must be of adequate thickness and have a marginal design that will allow it to bear the forces of mastication without fracture or deformation. In that regard, the restoration must have adequate occlusogingival depth to resist fracture in function or parafunction (bruxing or clenching). Second, the remaining tooth structure must be left in such a state that it, too, will resist the forces of mastication. As much sound tooth structure as possible must be maintained. If adequate resistance form cannot be maintained in the tooth to resist masticatory

forces, the weak portion of the occlusal surface should be cut away and replaced with amalgam or another strong restorative material.[64]

To maximize resistance form for tooth structure, minimum sound tooth structure should be removed when teeth are prepared for Class 1 or Class 2 amalgam restorations. Several studies[65–69] have demonstrated that, as an amalgam restoration becomes wider faciolingually, the tooth is more subject to fracture and the integrity of the restoration is less likely to be maintained. In Class 2 restorations, an increase in the depth of the occlusal portion of an amalgam restoration has also been linked to a decrease in resistance to fracture of the tooth.[66] Class 2 restorations that are confined to the marginal ridge areas (proximal slot restorations) may minimize the severity of tooth fracture compared with Class 2 restorations that extend through occlusal grooves.[70,71] Based on this knowledge, the following goals should guide the preparation and restoration of teeth: *(1)* removal of carious tooth structure, *(2)* preservation of the integrity of the tooth and periodontium, and *(3)* maximization of the life of the restored tooth.[60,72]

Class 1 Preparation

Indications

Occlusal caries

The indication for an initial Class 1 amalgam restoration is carious tooth structure in the occlusal fissures (or in facial or lingual pits in posterior teeth) detected clinically and confirmed with bitewing radiographs. The objectives of treatment are to eliminate caries lesions, to remove any enamel that has been undermined by the caries process, to preserve as much sound tooth structure as possible, and to create a strong restoration that mimics the original sound tooth structure and allows little or no marginal leakage.

For the purpose of this chapter, it is necessary to review the definitions of the terms *groove*, *fissure*, and *pit*.[73] A *groove*, or a *developmental groove*, is a linear channel on the surface of a tooth, usually at the junction of dental lobes (cusps or ridges). A *fissure* is a developmental linear cleft, the result of incomplete fusion of the enamel of adjoining dental lobes. A *pit* is a pinpoint fissure or the junction of several fissures.

The presence of deep or stained fissures alone does not justify placement of a restoration. When there is concern that dentin at the base of a deep fissure may become carious, the fissure should be sealed with a resin fissure sealant or flowable resin composite material. In a tooth that has been determined to have a localized fissure caries lesion or lesions, the carious tooth structure should be removed and one or more occlusal restorations placed, with the number of restorations determined by the number of areas in which carious tooth structure was removed. Remaining fissures that are considered to be susceptible to caries should then be sealed with a resin sealant (Fig 12-2).

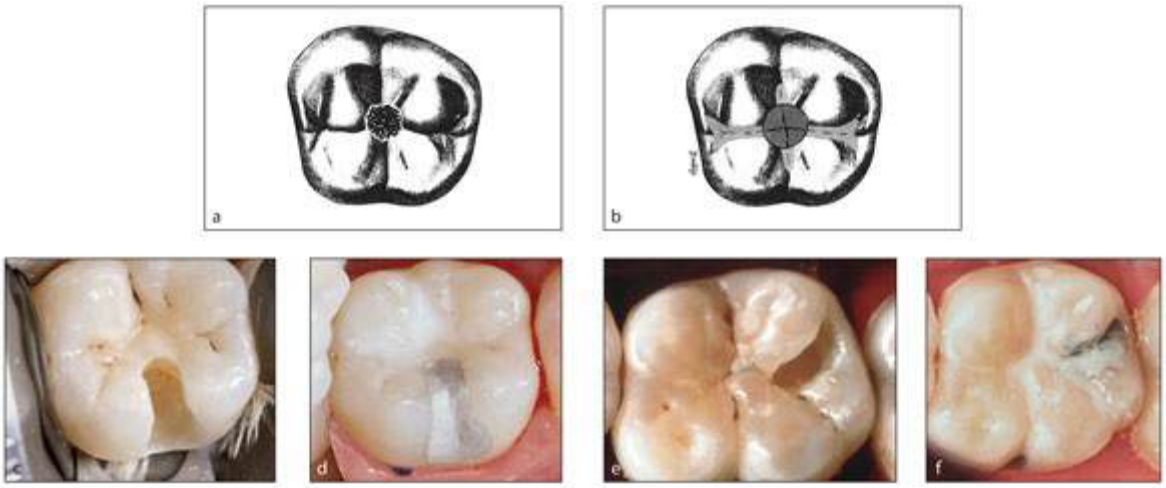


Fig 12-2 Amalgam should be used as a restorative material only in areas where actual carious tooth structure has been removed; remaining noncarious fissures and amalgam restoration margins are etched and sealed with resin fissure sealant. *(a)* This drawing shows the extent of carious dentin as if it could be seen through the occlusal enamel. *(b)* An amalgam restoration has been placed in the area where carious dentin and unsupported enamel were removed; the remaining fissures were sealed. *(c and d)* Occlusofacial preparation in mandibular molar; restoration with remaining occlusal fissures sealed. *(e and f)* Mandibular first molar preparation, sealed restoration, and sealed fissures.

If deep fissures that are to be sealed exhibit enamel demineralization or heavy stains, they may benefit from being prepared with a small bur (no. 1/16 or 1/8) to a width and depth of approximately 0.4 mm before being acid etched and sealed with a resin fissure sealant[74–76] (Fig 12-3). Alternatively, these fissures may be opened or cleaned with an air-abrasive instrument or an air-polishing prophylaxis unit[77–81] (Fig 12-4).

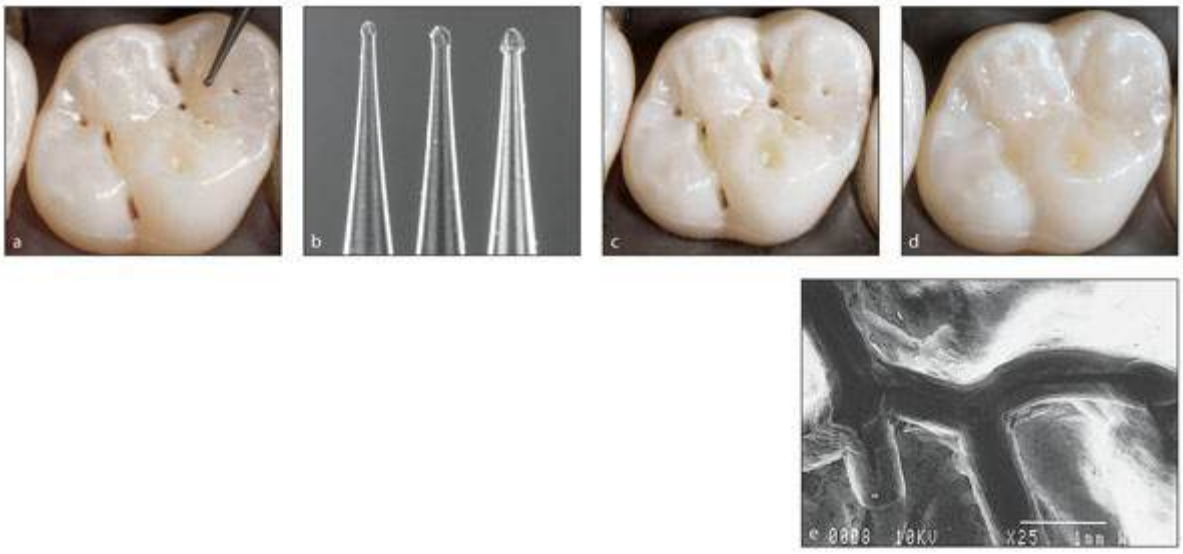


Fig 12-3 Fissures can be opened for sealing with a small round bur. *(a)* The fissure system of a maxillary first molar is shown before treatment with a no. 1/16 bur. *(b)* Any of three small round burs—nos. 1/16 bur (0.3- mm diameter), ⅛ bur (0.4-mm diameter), and ¼ bur (0.5-mm diameter)—may be used to open the fissures. *(c)* The fissures are prepared to the depth of the bur head. *(d)* The posttreatment occlusal surface shows the sealed fissures. *(e)* Scanning electron photomicrograph of fissures prepared with a small round bur.

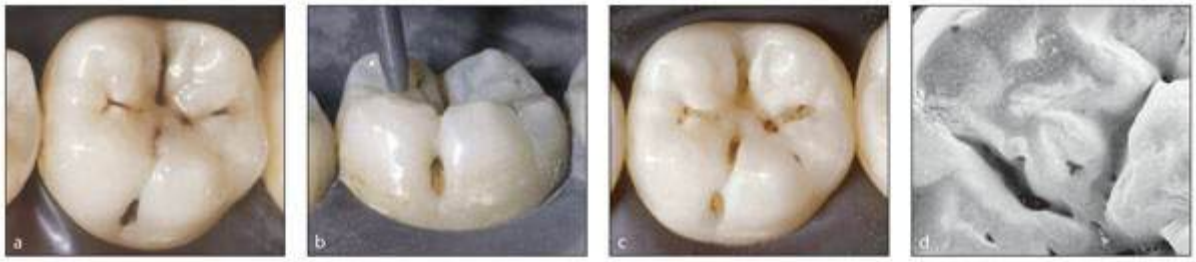


Fig 12-4 Fissures can be opened with an air-abrasion unit (KCP FlexiJet, American Medical Technologies). *(a)* The fissure system of a mandibular molar is shown prior to air abrasion. *(b)* 50-µm alumina powder is projected into the fissure. *(c)* The fissures are shown after air abrasion. *(d)* Scanning electron photomicrograph of fissures opened with air abrasion.

Traditionally, in a Class 1 amalgam preparation, occlusal fissures, or at least those in the developmental grooves, have been included in the preparation even when areas of the fissure system have not been carious. The justification for this has been that carious dentin, although not evident visually or radiographically, may be lurking at the base of one of those fissures. There is strong evidence that carious dentin inadvertently left at the base of a sealed fissure does not progress[82–86] and that the sealing of fissures associated with occlusal amalgam restorations is an extremely effective treatment.[87–90] Therefore, the routine extension of cavity preparations through fissures not known to be carious cannot be justified. Additionally, extension of cavity preparations through grooves in which there are no

fissures is contraindicated.

Defective restorations and recurrent caries

Another indication for a Class 1 restoration is the replacement of a restoration that is defective beyond repair or associated with a recurrent caries lesion. A *recurrent caries lesion* is one that occurs adjacent to an existing restoration. Most restorations exhibit some leakage at their margins, although it is minimal in most cases. When the leakage becomes greater, usually because of a defective restoration or flexure of tooth structure or the restoration, plaque can form in the space between the tooth and the restoration, which can produce demineralization to form a caries lesion. There is some evidence that cleaning, etching, and sealing leaking amalgam margins with a resin sealant can prolong the life of a restoration.[91,92]

Outline form

When an occlusal restoration must be placed because of initial caries lesions, two guidelines should be applied in establishing the outline form: *(1)* carious tooth structure should be eliminated, and *(2)* margins should be placed on sound tooth structure. The enamel at the margin of the preparation should be supported by sound dentin. Any enamel that has been undermined by the removal of carious dentin should be removed. If a noncarious fissure is evident in the wall of a preparation, the preparation should not be extended solely to include the fissure; the fissure should instead be sealed after the amalgam has been placed.

If there is no cavitation in the area of the lesion, a bur such as a no. 329 or 330 is used to cut through the enamel to gain access to the carious dentin. The preparation is widened to give access to all carious dentin and to remove any unsupported enamel. The preparation should be widened only enough to obtain enamel walls supported by sound dentin.

Although the outline form should not contain sharp angles, sound tooth structure should not be removed simply to obtain wide, smooth curves in the outline form. The outline form should be smooth to facilitate the uncovering of the margins during carving of amalgam. If, after the amalgam is carved, the margins of the preparation are jagged or rough, it is difficult for the dentist to know if this is because the enamel margin is rough or because amalgam is extending past the margins onto the surface of the tooth (overextended amalgam or amalgam flash).

When a defective restoration or a restoration associated with a recurrent caries lesion is to be replaced, the outline form will be determined by several factors.

First, the outline form of the old restoration will have a major influence. Also, the outline form may have to be extended because of additional pathosis. Finally, the resistance form for the tooth structure or restoration may have to be improved, and that will affect the outline form.

Resistance and retention form

To provide retention form for the amalgam, opposing walls of Class 1 occlusal restorations should be parallel to each other or should converge slightly occlusally. Enamel rods in most areas of the occlusal surface are directed roughly parallel to the long axis of the tooth and are approximately perpendicular to the tooth surface,[93] a factor that should be considered when the angulation of the margin of the amalgam preparation is designed. To enhance their ability to resist fracture, enamel margins should be prepared at a right angle or slightly obtuse angle (90 degrees or greater), as enamel margins of less than 90 degrees are more subject to fracture, especially in function (Fig 12-5). Clinical research has shown that clinical survival of amalgam restorations is inversely related to the degree of unsupported enamel.[94]

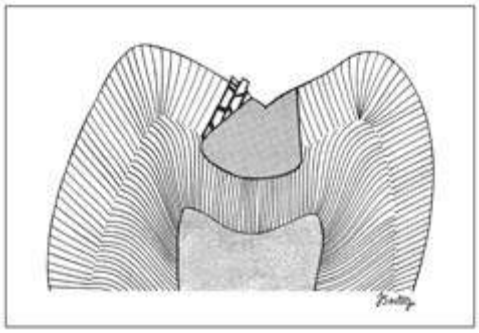


Fig 12-5 An acute cavosurface margin of enamel has the potential for fracture; a 90-degree enamel margin on the occlusal surface will withstand occlusion.

For resistance form in the amalgam restoration, amalgam margins should be approximately 90 degrees. Although many amalgam restorations will have amalgam margins that are significantly less than 90 degrees on the occlusal surface, very acute amalgam margins are much more subject to fracture (Fig 12-6). Marginal fracture will usually cause marginal gaps, or ditches, between the amalgam and the enamel.

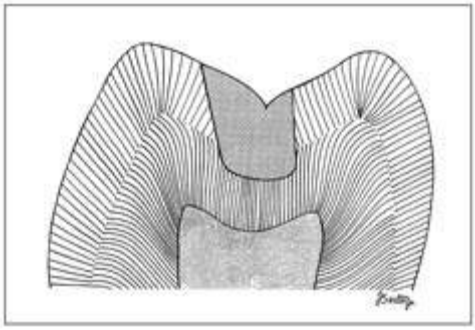


Fig 12-6 The left margin exhibits an acute "fin" of amalgam, which has a greater propensity for fracture, depending on the load applied to it during mastication. The right marginal configuration allows nearly a 90-degree angle for amalgam, imparting greater strength.

If the faciolingual width of the preparation exceeds one-third of the distance between the tips of the facial and lingual cusps (intercuspal distance), the remaining cusps themselves should be carefully evaluated. Even in narrower preparations, cusps should be evaluated for cracks that could lead to fracture, and the functional loading to which they will be exposed should be assessed. If a cusp is too weak to withstand function (Fig 12-7), it should be reduced for coverage with amalgam or attached in some way to the amalgam to provide cuspal reinforcement (described in the section on complex amalgam restorations).



Fig 12-7*(a)* The mesio-occlusodistal preparation in the premolar, after removal of a defective restoration, is greater than one-third the intercuspal distance; therefore, the cusps ideally should be protected by reduction and coverage with a restoration, or they should be cross-splinted (see Fig 12-39). *(b)* The mesio-occlusofacial preparation in the molar leaves facial cusps too thin to resist occlusal loading; they should be reduced and protected with a cuspal-coverage restoration. *(c)* The lingual cusp of the maxillary premolar, which was not protected or reinforced during restoration, has fractured.

It is generally accepted that occlusal amalgam restorations should have an occlusogingival thickness of at least 1.5 mm, and preferably 2.0 mm, to resist fracture during function (resistance form for the restoration). When carious dentin and the overlying enamel are removed, the preparation will be at least that depth and usually deeper.

Figure 12-8a shows a diagram of a cross section of the crown of a posterior tooth

with carious dentin at the base of the fissure. Figure 12-8b shows a cross section of the amalgam restoration that is indicated because of that caries lesion.

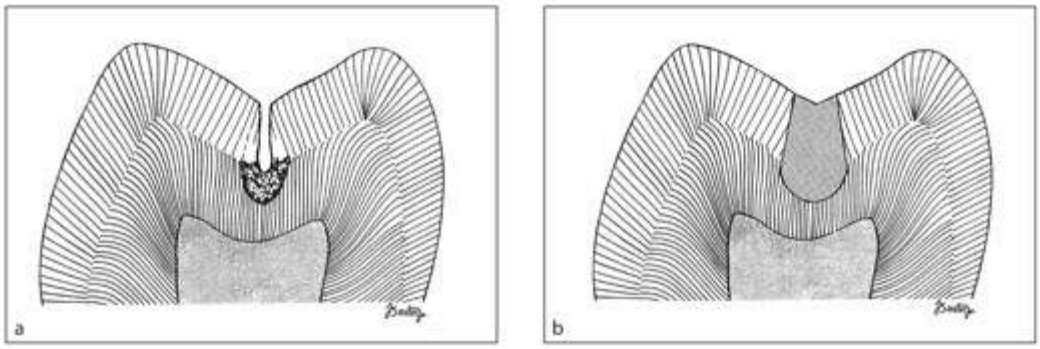


Fig 12-8*(a)* A fissure caries lesion in this tooth involves demineralization of the enamel at the depth of the fissure and advance of the lesion into dentin. *(b)* Tooth preparation involves removal of the carious dentin and of the enamel not supported by sound dentin. Only the diseased dentin is removed; additional dentin is not removed simply to create a flat pulpal floor for the preparation.

Figure 12-9 shows the outline form of several occlusal amalgam restorations; the outline form was determined by the extent of the demineralized dentin at the base of the fissures. Again, fissures not known to be carious, in a surface receiving an amalgam restoration due to fissure caries, should be sealed with a resin fissure sealant.

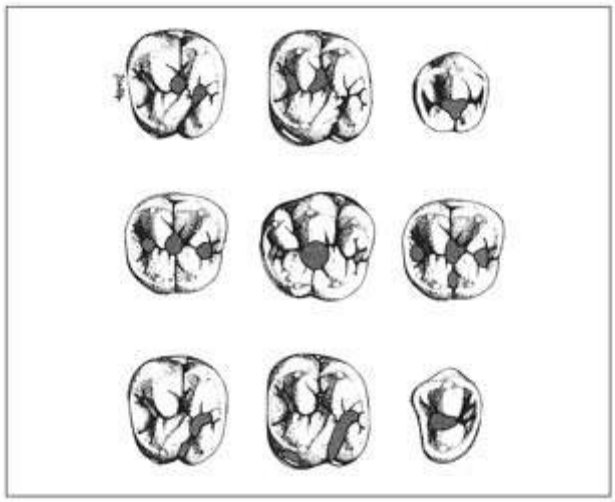


Fig 12-9 Outline form of various Class 1 preparations. The extent of carious demineralization is the determinant of outline form, so outline form will vary in every situation involving caries. Fissures not known to be carious should be sealed with a resin fissure sealant; fissures may be opened with a small bur to a depth of approximately 0.4 mm prior to sealing to ensure sound enamel for bonding.

If an occlusal caries lesion encroaches on the enamel of the proximal surface so that, when the carious dentin is removed, the proximal enamel has no dentinal support, consideration should be given to converting the Class 1 preparation to a

Class 2 preparation. An important part of this consideration should be the determination of the forces to which the marginal ridge will be exposed. If there is direct occlusal contact between the opposing tooth and the weakened marginal ridge, the marginal ridge should be removed and replaced with amalgam.

Class 2 Preparation

Indications

An initial Class 2 restoration is usually placed because a caries lesion is present in a proximal surface of a molar or premolar. Proximal caries lesions can sometimes be detected visually during a clinical examination, but they are usually detected with bitewing radiographs. The depth of the penetration of demineralization in enamel and dentin is actually greater than it appears to be in a bitewing radiograph. A caries lesion that appears radiographically to have penetrated about two-thirds of the way through the proximal enamel has actually penetrated the dentinoenamel junction (DEJ). However, even if the lesion has slightly penetrated the DEJ, the tooth still has the potential for remineralization if the etiologic conditions are changed[95–97] (see chapter 5). Each patient must be individually evaluated for improved oral hygiene, alteration of diet, and reduction in cariogenic bacteria before the decision is made to surgically remove a minimally deep caries lesion. In most cases, a restorative procedure should not be undertaken to treat a proximal caries lesion unless there is radiographic evidence of at least slight penetration of the lesion into dentin toward the pulp.[96]

Outline form

As with Class 1 restorations, Class 2 restorations that leave as much sound tooth structure as possible will contribute to resistance to fracture of the tooth. Tooth preparation necessitated by a caries lesion on a proximal surface should, when possible, avoid extension of the occlusal outline more than is necessary to allow access to the proximal lesion, to remove demineralized enamel and dentin, and to remove enamel not supported by sound dentin. If an occlusal caries lesion is present, it should be treated with a separate occlusal restoration. If the preparation necessitated by the occlusal caries lesion is in close proximity to the occlusal outline of the proximal preparation so that there is minimal or no sound tooth structure separating the two preparations, they should be joined. As when Class 1

occlusal restorations are placed, fissures not known to be carious but believed to be susceptible to caries should be sealed with a resin fissure sealant.[90] Fissures that contact the outline of a Class 1 or Class 2 preparation should be sealed. Retention form for the proximal restoration should be attained within the proximal preparation; the preparation should not be extended further into a sound occlusal surface to provide retention of the proximal restoration, because this will weaken the tooth's resistance to fracture.

Access to the proximal caries lesion is usually made by preparation through the marginal ridge (Fig 12-10). The proximal preparation begins with the creation of a slot, cut with a small bur in the center (mesiodistally) of the crest of the marginal ridge (Fig 12-10a) and occlusal to the proximal surface caries lesion (usually located faciolingually in the center of the interproximal contact area). The slot is deepened gingivally until the bur "falls" into the soft carious dentin, which has little resistance to the advance of the bur. The preparation is widened facially and lingually to eliminate all demineralized tooth structure at the DEJ and to remove enamel that is not supported by sound dentin. Demineralized enamel should usually be removed as well. However, if demineralization is superficial (less than half-way through the enamel) and there is evidence that the patient will reduce his or her caries risk status, consideration should be given to stopping the extension of the preparation short of removing some superficial demineralized enamel. After the amalgam restoration has been placed, the demineralized enamel can be treated with fluoride to enhance remineralization, or it can be acid etched and coated with light-cured resin for at least short-term protection from further demineralization.[98] Alternatively, a low-viscosity resin infiltrant (eg, Icon, DMG) can be applied to the demineralized enamel following acid etching to inhibit further demineralization.[99]

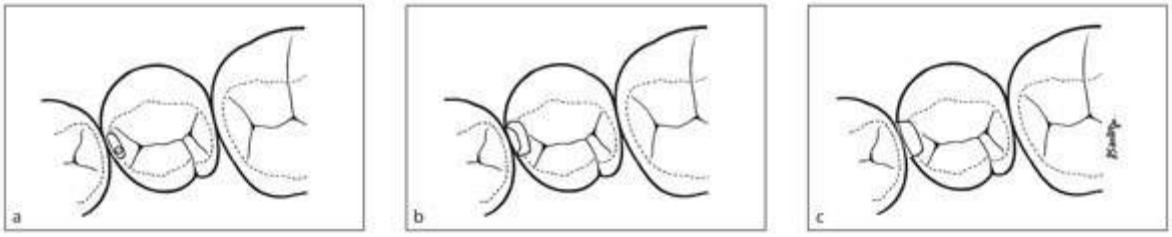


Fig 12-10*(a)* For a Class 2 preparation to treat an initial proximal surface caries lesion, an initial cut is made through the marginal ridge with a narrow bur to penetrate to carious dentin; then the slot is widened faciolingually. *(b)* The mesial and distal aspects of the proximal enamel plate are thinned to facilitate its fracture and removal. *(c)* The final preparation will be determined after removal of all carious tooth structure and removal of fragile enamel (retention would be by amalgam bonding or the addition of retention grooves).

The Class 2 restoration necessitated only because of a proximal caries lesion and

having an occlusal outline limited to the marginal ridge area will be referred to in this chapter and others as a *slot restoration*. If it involves the distal surface with access from the occlusal surface, it will be called a *disto-occlusal slot restoration* (Fig 12-11).



Fig 12-11 Preparations for proximal slot restorations. Note significant retention grooves. Such mechanical retention should be used when the restoration will not be bonded.

When preparing the tooth for the proximal slot restoration, a shell of enamel should be left between the preparation and the adjacent tooth (Figs 12-10b and 12-12). This will prevent accidental nicking, scarring, or other damage to the adjacent tooth. One study[100] found that proximal surfaces of adjacent teeth were damaged during Class 2 preparation 69% of the time and that these damaged surfaces were almost three times as likely to become carious as were undamaged surfaces. Special care to avoid damage to the adjacent tooth is warranted. Any nicking or scarring of an adjacent tooth should be polished away before the restoration is placed.

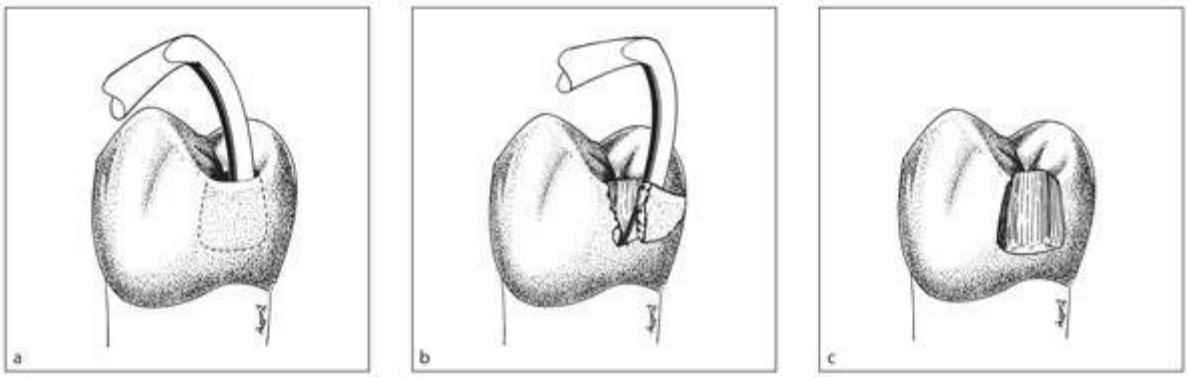


Fig 12-12*(a)* The plate of proximal enamel should be fractured with a hand instrument to prevent damage to the adjacent tooth by a bur. The instrument is placed in the slot. *(b)* The instrument is rotated to fracture the plate. *(c)* The fractured margins have been planed with a gingival margin trimmer or hatchet.

After the removal of carious tooth structure and fragile enamel, the proximal surface margins of a Class 2 amalgam preparation will not usually be in contact with the adjacent tooth (Fig 12-10c). However, if all demineralized and undermined

enamel has been removed and a margin remains in contact with the adjacent tooth, consideration should be given to allowing contact to remain (Fig 12-13).



Fig 12-13 Class 2 proximal slot restoration showing the lingual cavosurface margin remaining in contact with the adjacent tooth after removal of all carious tooth structure and unsupported enamel.

During the removal of carious dentin, the demineralized dentin in the periphery of the preparation (at or near the DEJ) should be removed and the outline form extended to ensure that the enamel at the margins of the preparation is supported by sound dentin. Carious dentin should be removed with the largest round bur that will fit into the area. After the periphery of the preparation is clear of demineralized tooth structure, the carious dentin near the pulp should be removed. The bur should be rotated very slowly in a low-speed handpiece; the rotation should be so slow that the individual blades of the bur can be seen as it rotates. The blades of the slowly rotating bur are like multiple spoon excavator blades, but the depth that a blade can penetrate into the carious dentin is limited by the edge angle of the bur and by the depth of each bur blade toward the center of the bur, so the bur will remove only a limited depth of carious dentin during each rotation. During removal of deep carious dentin, this procedure is less likely to result in a pulpal exposure than the use of a spoon excavator.

Resistance and retention form

After the shape of the preparation is roughed out with a bur, hand instruments (eg, a gingival margin trimmer) may be used to fracture away the shell of enamel; to shape the facial, lingual, and gingival walls and margins; and to scrape away any fragile enamel from the margins (Fig 12-14). The facial and lingual walls of a Class 2 slot preparation should converge slightly toward the occlusal surface to provide retention form for the restoration (Fig 12-15).

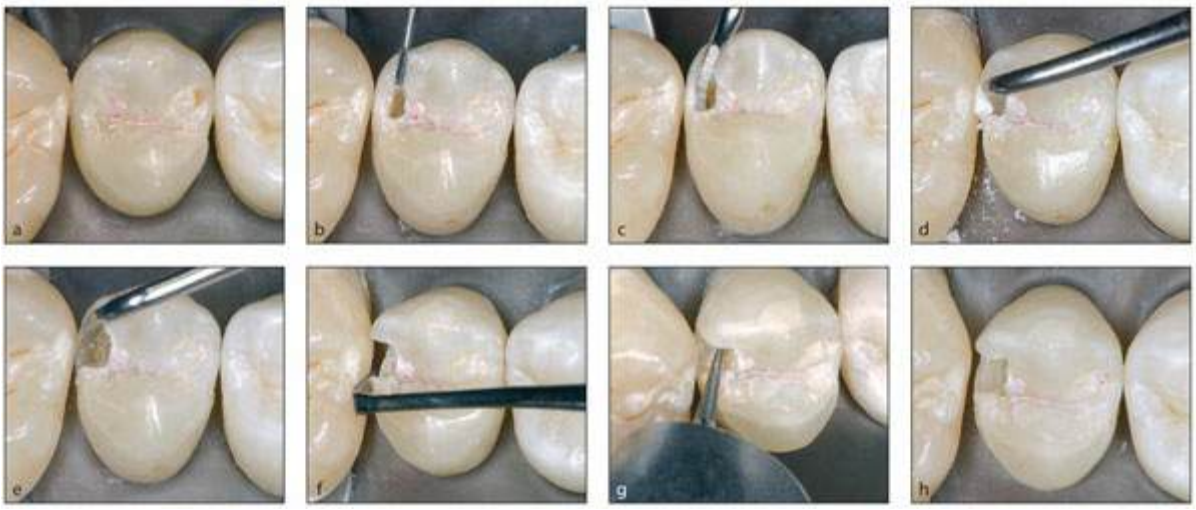


Fig 12-14*(a)* A disto-occlusal slot amalgam preparation will be made because an initial caries lesion is present. *(b)* A small bur is used to cut a slot, beginning at the crest of the marginal ridge and extending gingivally to "fall" into carious tooth structure. A thin plate of enamel separates the bur from the adjacent tooth to prevent damage to the adjacent tooth. *(c)* A gingival margin trimmer is placed into the slot created by the bur and rotated to apply pressure to the thin plate of proximal enamel. *(d)* The thin plate is fractured. *(e)* The walls and margins are planed with the margin trimmer to smooth them and to eliminate any remaining fragile enamel. *(f)* When there is some separation of the proximal margin from the adjacent tooth, a thin carver, such as this interproximal carver, may be used to carve and burnish the amalgam. If a portion of a proximal margin is in contact with the adjacent tooth, the carver may be used to carve and burnish the amalgam occlusal and gingival to the area of contact. *(g)* Retention grooves are placed with a no. ⅛ or ¼ bur. *(h)* The preparation is completed.

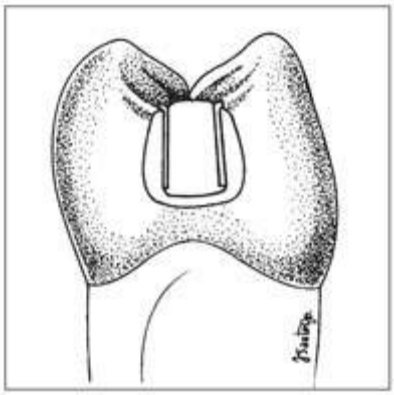


Fig 12-15 The proximal slot preparation, or the proximal box of a Class 2 preparation, should have walls that meet the proximal surface of the tooth at 90 degrees and converge occlusally to provide resistance and retention form.

To provide resistance form for the Class 2 amalgam restoration, the proximal preparation should have a mesiodistal dimension of about 1.5 mm or more. If there is sound dentin supporting occlusal enamel in the fossa adjacent to the marginal ridge, that dentin and enamel should be left intact. If the caries lesion extends from the proximal DEJ deeper into dentin, the demineralized dentin should be removed

completely, especially in the areas near the DEJ, and sound dentin should be left in place.

The gingival floor of the proximal preparation may be flat and approximately perpendicular to the long axis of the tooth, or it may be curved and/or slanted faciolingually, as determined by the extent and configuration of the caries lesion that necessitated the restoration. The location of the gingival floor, therefore, should be determined by the gingival extent of the carious lesion. The gingival wall, like the facial and lingual walls of the proximal preparation, should form an angle of approximately 90 degrees with the surface of the tooth. This provides strength to both the amalgam and the enamel and prevents enamel not supported by sound dentin from being left at the margins of the restoration.

Convergence toward the occlusal surface of the facial and lingual walls of the proximal slot preparation gives retention form to the restoration to keep it from dislodging occlusally. Although with initial proximal surface caries lesions it is not often necessary to extend the Class 2 preparation into occlusal grooves, the operator will frequently need to replace an existing restoration that was prepared with an occlusal extension. If the restoration is extended into occlusal grooves, this extension will provide resistance to displacement of the restoration proximally (ie, toward the adjacent tooth). To provide enough resistance, however, the extension into the occlusal surface must have a faciolingual dimension of at least one-fourth the distance between the facial and lingual cusp tips[101] (*intercuspal distance*), and the facial and lingual margins of the occlusal extension must be approximately parallel to each other in a mesiodistal direction (Fig 12-16a).

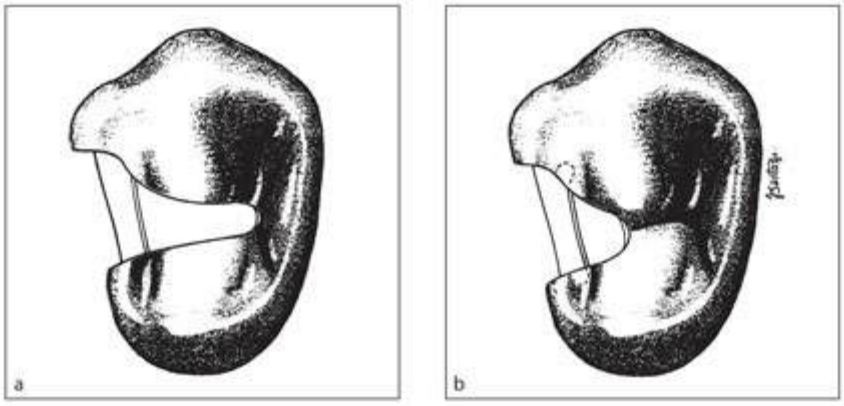


Fig 12-16*(a)* Although extension of a Class 2 preparation into occlusal grooves is not usually necessitated by carious tooth structure, if such an extension is already present from a previous restoration of the tooth, it will provide some resistance form for the proximal portion. Note the slight bevel of the axiopulpal line angle as part of the resistance form. *(b)* An extension without parallel cavosurface margins, however, will not provide the resistance form needed to prevent displacement of the proximal amalgam during mastication, so retention grooves *(dotted lines)* should be added.

Undercuts

If the extension into the occlusal surface is narrower, if there is no extension into the occlusal surface, as with the proximal slot restoration, or if no amalgam bonding system is to be used, *retentive undercuts* (retention grooves or points) must be cut into the dentin of the facial and lingual walls of the proximal box (Fig 12-16b). Use of a no. ¼ (ISO 005) round bur, with a head diameter of 0.5 mm,[101–103] or a no. ⅛ (ISO 004) round bur, with a head diameter of 0.4 mm, is recommended for preparation of retention grooves and points for Class 2 amalgam restorations (see Fig 12-14g). For a proximal slot restoration, retentive undercuts should be very distinct (at least 0.5 mm deep) and should oppose each other to form a dovetail effect in the dentin. Long grooves, extending from the gingival floor to the occlusal surface, are recommended for a proximal slot restoration[103,104] (Fig 12-17a). When no amalgam bonding system is used, if the occlusal extension is narrow, short retention grooves or retentive points should be prepared in the dentin of the facial and lingual walls to supplement the resistance form provided by the occlusal extension[102] (Fig 12-17b). If there is a bulky extension of amalgam into the occlusal surface of the tooth, retentive undercuts should not be necessary[101] (Fig 12-17c). For the preparation of any retentive undercuts with a bur, it is advisable to use a handpiece at low speed, without water spray, and to use magnification to enable visualization, because the location and direction of the undercuts are critical to the success of the restoration.



Fig 12-17*(a)* In a proximal slot or proximal box–only preparation, retention grooves should be long and very distinct. *(b)* In preparations with narrow occlusal extensions, short (0.5- to 1.0-mm) retention grooves or points are placed in facial and lingual walls just gingival to the occlusal DEJ. *(c)* When the occlusal extension is wide (1.5 mm or more faciolingually) and has parallel walls, retention points or grooves are not necessary.

Retentive undercuts in the dentin of the facial and lingual walls should be completely in dentin and not at the proximal DEJ; this obviates the removal of the dentinal support for the proximal enamel adjacent to the restoration. The undercuts

should not, however, be placed so far away from the DEJ that the pulp chamber could be penetrated. A good rule is to place a retentive undercut so that there is approximately 0.25 to 0.5 mm of dentin between the groove and the DEJ and so that the groove is approximately 0.5 mm deep and 0.5 mm wide. Again, the 0.5-mm diameter of the no. ¼ round bur, or the 0.4-mm diameter of the no. ⅛ bur, is ideal as a gauge of dimension. So that retentive undercuts do not penetrate through the dentin to the DEJ when they are placed in the facial and lingual walls, they should be cut parallel faciolingually to the DEJ and to the external surface of the tooth (Fig 12-18).

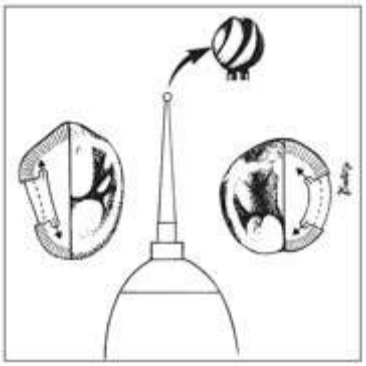


Fig 12-18 Location and direction of retentive undercuts (retention grooves) for the proximal portion of a Class 2 restoration. The illustrated grooves are approximately 0.5 mm wide and 0.5 mm deep (no. ¼ bur head is shown) and are directed approxi-mately parallel to the DEJ (and the external surface of the tooth). *(left)* For a fairly flat surface, such as the proximal surface of a maxillary premolar, the grooves are directed almost in the facial and lingual directions. *(right)* For a convex proximal surface, such as in a mandibular premolar, their direction is considerably vectored.

When retentive undercuts are necessary, they must be actual undercuts in the facial and lingual dentin that oppose each other (see Fig 12-18). This is especially important in proximal slot restorations if amalgam bonding is not used, because the undercuts are the only feature that will prevent dislodgment of the restoration proximally. In the preparation with a deep proximal box, the retention grooves should be in the proximal walls just inside the DEJ and not in the corners of the box (Fig 12-19); this is to reduce the risk of pulp exposure. Without correctly located, distinct retention grooves, a proximal slot restoration is more likely to become dislodged.

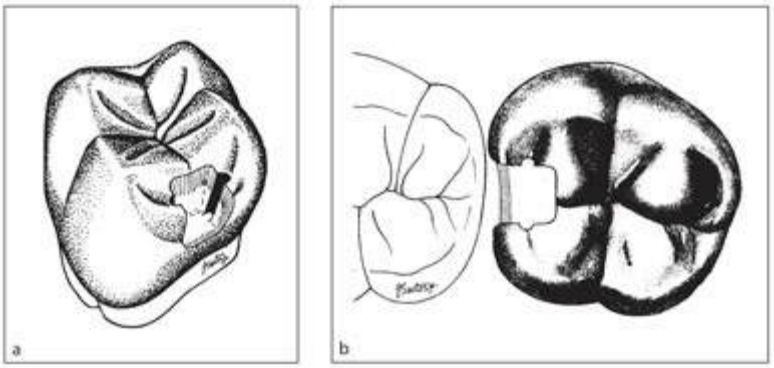


Fig 12-19 When, because of the axial depth of the carious dentin at the occlusal DEJ, the axial wall in a Class 2 preparation must be deeper pulpally than approximately 1 mm inside the DEJ, retention grooves (or points) should be placed so that they are just inside the proximal DEJ, not at the line angles of the deep preparation. *(a)* Proximal view of an axially deep Class 2 preparation showing retention grooves in dentin near the proximal DEJ and extending occlusally through the occlusal enamel; *(b)* occlusal view.

When well designed, the proximal slot restoration can last indefinitely. Figure 12-20 shows a Class 2 slot amalgam restoration that had been in function for more than 58 years.



Fig 12-20 The mesio-occlusal slot restoration in the maxil-lary first molar, placed by the late Dr Miles Markley of Denver, Colorado, had been in service for more than 58 years when this photograph was taken.

Mechanical retention

If a proximal box or slot is so wide that retentive undercuts will not oppose each other (Fig 12-21a), another type of retention and resistance method, such as amalgam bonding or a self-threading pin, placed horizontally or vertically, should be used (Fig 12-21b). Amalgam bonding and the use of pins are discussed later in this chapter.

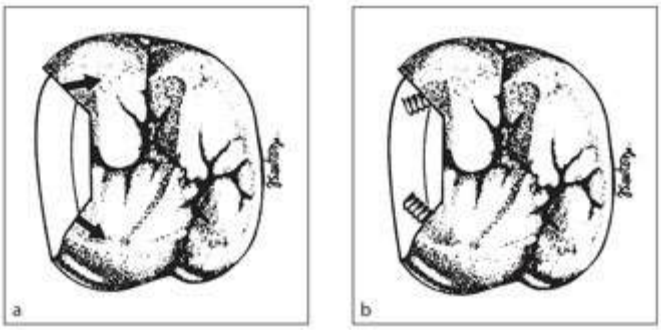


Fig 12-21 *(a)* Undercuts (retention grooves) should not be placed for a proximal box so wide that those undercuts, if placed parallel to the adjacent DEJ, would not oppose each other to provide needed undercut retention. *(b)* Instead, other types of resistance and retention features, such as these horizontal self-threading pins, should be employed.

Because the outline form of Class 2 amalgam restorations is always determined by the pathologic problem being treated, there are an infinite number of variations in occlusal outline form. Figure 12-22 illustrates the outline form of some typical Class 2 restorations that would be placed to treat initial caries lesions.

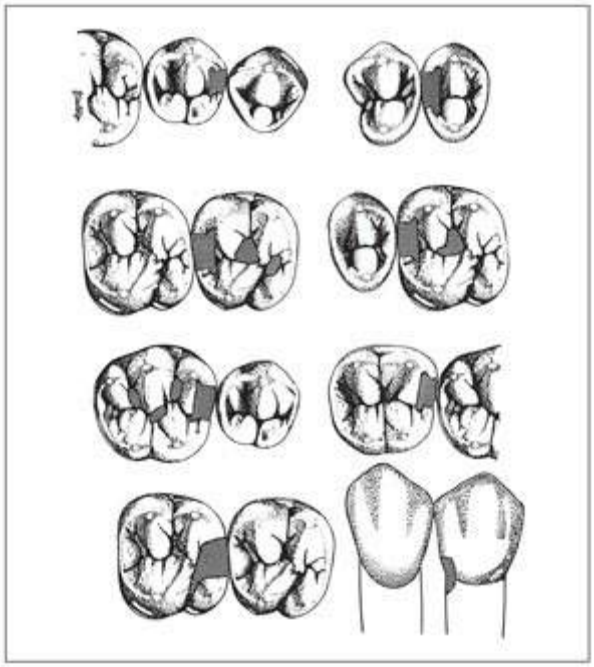


Fig 12-22 Typical outline forms of Class 2 restorations placed to treat initial caries lesions. Because outline form is determined by the pathosis present and the morphology and position of the tooth, there are an infinite number of variations. The bottom right-hand restoration represents a facial slot or keyhole restoration to treat root caries or caries at the CEJ.

Facial and lingual access

Most Class 2 amalgam restorations have occlusal access, as already discussed. If a proximal caries lesion is at or apical to the cementoenamel junction (CEJ), however,

it is often more conservative to use facial, and occasionally lingual, access.[105] The preparation for a Class 2 amalgam restoration with facial or lingual access, sometimes referred to as a *keyhole preparation* or a *facial* or *lingual slot preparation*, is similar to a slot preparation with occlusal access. The entire preparation is apical to the proximal contact. The location and configuration of the carious tooth structure and the access preparation determine the outline form of the keyhole preparation (see Fig 12-22, bottom right). Its margins, which are at least partly on cementum or dentin, should provide for cavosurface angles of approximately 90 degrees. Either amalgam bonding or groove retention, similar to that recommended for a proximal slot preparation with occlusal access, is indicated. Groove retention can usually be placed with a no. ¼ or ⅛ bur in the dentin of the occlusal and gingival walls; occasionally, grooves must be placed with a hand instrument, such as a gingival margin trimmer.

Replacement of restorations with occlusal extensions

Patients commonly will have existing Class 2 amalgam restorations that have extensions of the preparation through occlusal fissures and even through nonfissured grooves. Although this preparation outline form should not be advocated, existing restorations of this type will need to be replaced from time to time. Therefore, some design concepts to promote their longevity must be discussed. First, the narrower (faciolingually) the extension through the occlusal surface, the less marginal breakdown will occur[65,68] (Fig 12-23). Second, the junction of the proximal portion and the occlusal extension of the Class 2 amalgam preparation must have adequate depth (1.5 to 2.0 mm). The occlusal outline form at the junction of the proximal and occlusal components should not be sharp or jagged, but no more than a very small amount of sound cuspal enamel should be sacrificed to make the junction slightly rounded instead of sharp. It is generally accepted that, if there is an axiopulpal line angle at minimal occlusal depth, it should not be sharp; if there is such a sharp line angle, it should be rounded in order to reduce the concentration of stresses in that portion of the restoration.

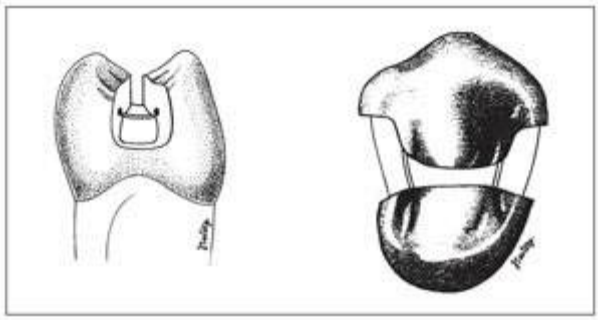


Fig 12-23 When extension through occlusal fissures or grooves cannot be avoided, such as when an old, defective amalgam restoration is being replaced, the faciolingual width of the occlusal portion of the preparation should be kept as narrow as possible to maintain tooth strength. *(left)* Proximal view; *(right)* occlusal view of mesio-occlusodistal preparation.

Complex Preparations

Historically, the term *complex amalgam restoration* referred to one that involved three or more surfaces of a tooth. The term has been redefined in recent years[61] to refer to an amalgam restoration that replaces one or more cusps.

When a cuspal-coverage restoration is indicated, a gold alloy casting is considered by many to be the restoration of choice. Gold has wear characteristics similar to those of enamel and has the ability to maintain a stable occlusion. Newer ceramic materials with high strength and wear resistance also represent viable options for these restorations. However, for various reasons, an indirect material cannot always be chosen as the definitive restoration. In these situations, amalgam is an excellent alternative restorative material.

The efficacy of the amalgam cuspal-coverage restoration has been shown in both laboratory and clinical studies.[61,62,106,107] The key to the successful placement of cuspal-coverage restorations is a thorough understanding of the underlying engineering principles. Preparations for amalgam restorations have traditionally been designed to provide adequate retention form. *Retention* has been defined as prevention of dislodgment of the restoration along the path of insertion (with tensile forces). *Resistance* is defined as prevention of dislodgment or fracture by oblique or compressive forces. Although retention form is important in the complex amalgam restoration, more emphasis should be placed on the resistance of both the restoration and the remaining tooth structure. Retention and resistance form can be obtained through the use of metal threaded pins, nonpin mechanical features, and amalgam bonding, all of which will be described.

Cuspal-coverage preparations

Often, individuals seek treatment because of a fractured cusp or cusps in posterior teeth. If the treatment option agreed upon for the tooth is a complex amalgam restoration, the tooth preparation will usually include removal of any existing restoration, removal of any carious tooth structure and fragile enamel and/or dentin, and preparation of margins to provide a cavosurface angle of approximately 90 degrees in all areas. In addition, weak cusps that have not fractured should be reduced for coverage and protection with amalgam.

The thickness of amalgam needed for cuspal protection will vary depending on the functional load to which the cusp will be exposed. A good guideline for amalgam thickness in centric holding cusps (stamp or functional cusps) of molars and premolars is 2.5 mm. In a facial cusp of a maxillary premolar, on those occasions when there is little or no function on the facial cusp, a reduced thickness of amalgam is acceptable to allow a maximum amount of facial enamel to remain for esthetics (Fig 12-24).



Fig 12-24*(a)* Because of a diagnosis of incomplete tooth fracture, the maxillary premolar has been prepared for a complex amalgam restoration. *(b)* The tooth has been restored with a complex amalgam restoration that protects the entire occlusal surface. *(c)* The occlusal height of the mesial aspect of the nonfunctioning facial cusp was only reduced 1.5 mm to preserve facial enamel for esthetics.

When cusps are reduced for coverage, the occlusal tooth structure should be reduced anatomically to provide for an adequate and consistent occlusal amalgam thickness. To facilitate consistent reduction, depth cuts are recommended. Figure 12-25 illustrates the use of depth cuts to ensure consistent reduction of occlusal tooth structure for coverage. Figure 12-26 shows a clinical case in which depth cuts provided consistent reduction.

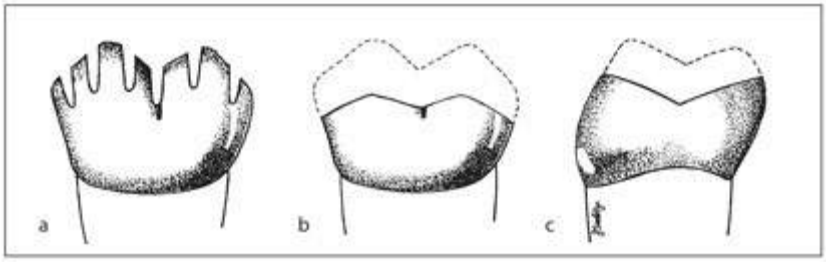


Fig 12-25 Depth cuts are used to provide for even reduction of occlusal tooth structure of a mandibular molar and consistent thickness of amalgam. *(a)* Depth cuts 2.5 mm deep; *(b)* cuspal reduction viewed from the facial aspect; *(c)* cuspal reduction viewed from the mesial aspect.



Fig 12-26 Reduction of weak cusps of a mandibular molar for coverage. *(a)* An instrument is placed so that it touches cusp tips of the adjacent teeth. A note can be made of the position of the cusps to be reduced so that they can be rebuilt in amalgam and carved to approximately the correct height before the rubber dam is removed. *(b)* Half the 5.5-mm length of a no. 169L bur head is used to make depth cuts approximately 2.5 mm deep in the cusps. *(c)* The depth cuts are completed. *(d)* The head of the handpiece is rotated so that the no. 169L bur can be used to reduce the cuspal structure between the depth cuts. *(e)* Facial cusps are reduced. *(f)* All cusps are reduced, and resistance features are placed. *(g)* Amalgam is placed, carved, and smoothed. The instrument is placed as it was prior to cusp reduction to ensure that cuspal height is similar to preoperative cuspal height. *(h)* Completed restoration. *(i)* Polished restoration.

The length of the head of the bur that is used for depth cuts must be known. Because head lengths vary from manufacturer to manufacturer, and even among burs of a single manufacturer, it is good practice to measure the length of a bur head prior

to preparation of depth cuts (Fig 12-27). A periodontal probe should be available for measuring the length of bur heads.

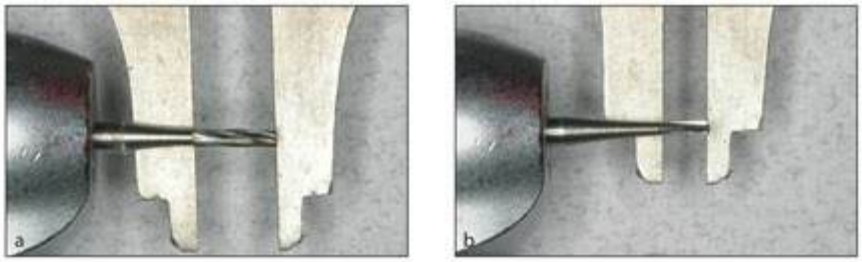


Fig 12-27*(a)* A Boley gauge is used to measure a no. 56 bur head (4 mm in this case). *(b)* A Boley gauge is used to measure a no. 330 bur head (2 mm in this case). The periodontal probe is another handy instrument for measuring bur head length.

Consistent reduction of cusps provides anatomical reduction rather than flat reduction. Anatomical reduction imparts adequate strength to the amalgam while preserving and protecting as much natural tooth structure as possible. Some operators reduce cusps to a flat surface perpendicular to the long axis of the tooth; no scientific justification for this practice can be found in the literature.

A timesaver in practice is to take note of cuspal height and cusp tip location or even to make a drawing or photograph prior to cuspal reduction so that cusps may be built and carved back to their original height prior to removal of the rubber dam (Fig 12-28; see also Figs 12-26a and 12-26g).

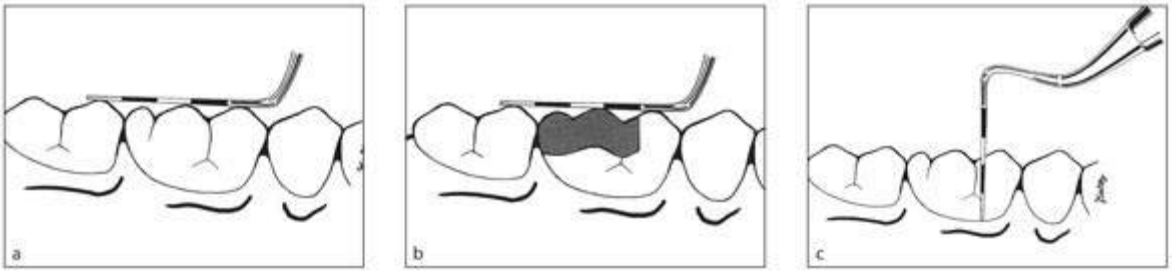


Fig 12-28 Preoperative registration of the height of cusps to be reduced and restored with amalgam. *(a)* The midfacial and distofacial cusps are to be reduced for coverage. A periodontal probe is placed along the facial cusp tips of the tooth to be restored and the adjacent teeth, and the relationships of the cusp tips to the probe are remembered or drawn. *(b)* The amalgam cusp tips of the carved restoration are seen to have a similar relationship to the probe. *(c)* If there are no adjacent teeth or cusp tips to guide the height of amalgam cusp tips, the distance from a landmark (such as the cervical line) may be measured with a periodontal probe.

Resistance and retention methods

For amalgam restorations that do not replace cusps, or at least large portions of cusps, the walls of the preparation provide retention and resistance form. Retention form is provided by convergent walls and undercuts placed in dentin. When a large

amount of cuspal tooth structure is lost or removed, the walls, or portions of them, which provide resistance and retention for the amalgam, are lost. For this reason, it is necessary to add features or adhesives to the preparation that will provide adequate resistance and retention for the restoration. Several methods of obtaining resistance and retention for complex amalgam restorations are discussed.

Pins

Although pins were first described in the 19th century,[108,109] Markley[110–112] popularized the concept of cemented stainless steel pins. Later, stainless steel pins, which were malleted into slightly undersized channels in dentin (friction-locked pins),[113] and threaded stainless steel pins, which were screwed into channels in dentin,[114] were developed. Laboratory studies have since investigated the properties of these three types of pins (cemented, friction-locked, and self-threading), and because of these studies, the self-threading pins are the primary pins in current use (Fig 12-29). In a study by Dilts et al,[115] self-threading pins were found to be more retentive in dentin than cemented or friction-locked pins. The authors also recommended a depth in dentin of 2 to 3 mm as optimum for self-threading pins. Moffa et al[116] found that a pin length of approximately 2 mm into amalgam provides optimum retention. The relationship between retention and the diameter of the pin has also been investigated. As would be expected, larger-diameter pins are more retentive.[115–117]

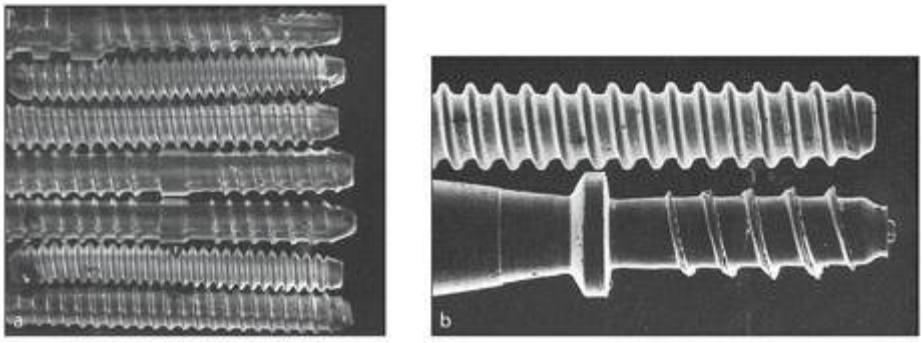


Fig 12-29 Scanning electron micrographs of self-threading pins. (Courtesy of John O. Burgess, University of Alabama at Birmingham School of Dentistry.) *(a) (top to bottom)* PPS (Brasseler; titanium alloy), Denlok (Denovo; stainless steel), Filpin (Ivoclar Vivadent; titanium), TMS (Thread Mate System) Link Plus (Coltène/Whaledent; titanium alloy), TMS Link Plus (Coltène/Whaledent; stainless steel), TMS Minim (Coltène/Whaledent; stainless steel), Stabilok (Fairfax Dental; stainless steel). *(b) (top)* TMS Minim pin; *(bottom)* Max pin (with threads more widely separated to allow for greater thickness of dentinal threads).

Self-threading pins have been manufactured in a variety of configurations. Some

are self-shearing and some have heads. Figure 12-29a shows several pins from various manufacturers. In one study of self-threading pins,[118] pins manufactured by Coltène/Whaledent demonstrated superior resistance and retention. Although several manufacturers have discontinued the production of retentive pins, those pins produced by Coltène/Whaledent are still available.

Currently marketed pins have the metal threads separated to provide thicker, bulkier dentinal threads. When a pin is pulled from a pin channel, it is the dentinal threads that shear and not the metal threads. The pin design with wider dentinal threads is retained well in dentin (Fig 12-29b). Another feature of many of the currently available pins is a shoulder stop. The purpose of this feature is to prevent the end of the pin from putting stress on the dentin at the end of the pin channel; the Max pin (Coltène/Whaledent) has an effective shoulder stop incorporated into its design. A shoulder is a part of the design of the Link Plus pin (Coltène/Whaledent), but its diameter is similar to that of the threads, so it does not provide an effective stop.[119] Although a definite shoulder stop is theoretically beneficial, there is no evidence of problems associated with pins that lack effective shoulder stops.

Coltène/Whaledent pins and pin channel (or twist) drills in Regular, Minim, and Minikin sizes are shown in Fig 12-30; a smaller size (Minuta, 0.0135-mm diameter pin channel drill; 0.015-mm diameter pin) is available, but there seems to be little practical use for it. The gold-plated stainless steel TMS Regular and Minim pins (see Figs 12-35b and 12-35c) are also available as self-shearing pins and in double-shear (two pins in one) form as well as single-shear form. All TMS Link and Link Plus (with shoulder) pins are self-shearing; Link Plus Regular and Minim pins are also available in the two-in-one (double-shear) form. The Link and Link Plus pins are available in either gold-plated stainless steel or titanium alloy; these may be inserted manually or with any low-speed, latch-type handpiece. The bulk TMS pins are available only in gold-plated stainless steel and are usually inserted manually. Selection among these pins should be based on operator preference; they all have performed well in laboratory studies.[118,120]

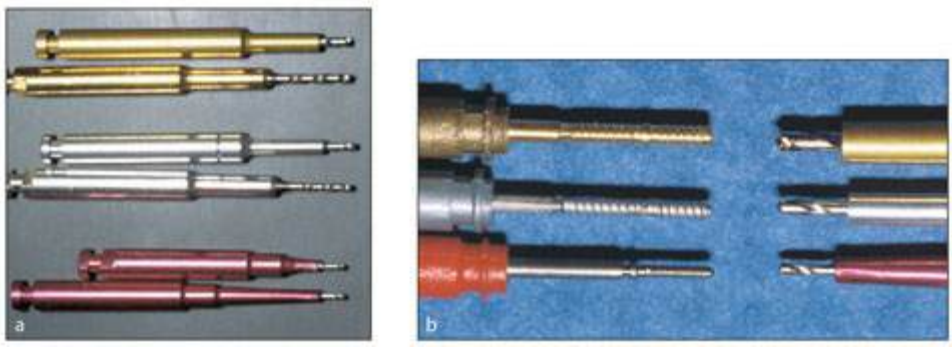


Fig 12-30 Color-coded pin channel (twist) drills and pins of various diameters and lengths

(Coltène/Whaledent TMS). *(a)* Pin channel (Kodex) drills: *(top to bottom)* Regular (gold, 0.027-inch [0.675-mm] diameter, 2.0- and 5.0-mm lengths); Minim (silver, 0.021-inch [0.525-mm] diameter, 2.0- and 5.0-mm lengths); Minikin (red, 0.017- inch [0.425-mm] diameter, 1.5-mm length). *(b)* Pins with corresponding pin channel drills: *(top to bottom)* Regular (0.031-inch diameter) gold-plated stainless steel Link Plus pin with Regular (0.027-inch diameter) pin channel drill (2.0-mm depth-limiting); Minim (0.024-inch diameter) titanium alloy Link Plus pin with Minim (0.021-inch diameter) pin channel drill (2.0-mm depth-limiting); Minikin (0.019-inch diameter) titanium alloy Link Series pin with Minikin (0.017-inch diameter) pin channel drill (1.5- mm depth-limiting).

Number of pins to use. It is difficult to develop a guideline that would determine the appropriate number of pins for all situations. It has been demonstrated that as the size and number of pins increase, the amount of resistance form imparted by the pins increases[121]; however, the number of pins used will vary with the size of pin, the amount of remaining tooth structure, other mechanical resistance features used, the use of amalgam bonding systems, and the expected functional requirements of the definitive restoration.

Channel preparation. A rubber dam should be in place when pin channels are prepared and when pins are placed to protect the patient from aspiration and to prevent contamination by saliva in case there is pulpal perforation during pin channel preparation.

Because the rotating tips of pin channel drills tend to move around on the cut dentin surface, it is usually helpful to place an indentation or starting point in the dentin at the desired location for the initiation of the pin channel. The starting point may be placed with a small bur, such as a no. ¼ or ⅛ bur.

Various lengths and diameters of pin channel drills are available for preparation of pin channels (see Fig 12-30). The most popular pin channel drills have depth-limiting shoulders, which ensure that the optimum pin channel depth is not exceeded. To avoid perforation of either the pulp or the external surface of the tooth, location of the pin channel is critical. The channel should usually be prepared parallel to the nearest external tooth surface. Before channel preparation is initiated, approximately 2 mm of the end of the pin channel drill should be placed against the external surface of the tooth. If that much of the side of the tooth is exposed above the rubber dam, alignment is facilitated. Frequently, however, adjacent soft tissue under the rubber dam obscures visualization of the tooth surface in the area adjacent to the desired pin location. For alignment, the drill is placed against the external tooth surface, and the angulation of the drill is changed until the drill separates from the margin of the preparation; it is then rotated back until it just contacts the margin (Fig 12-31).

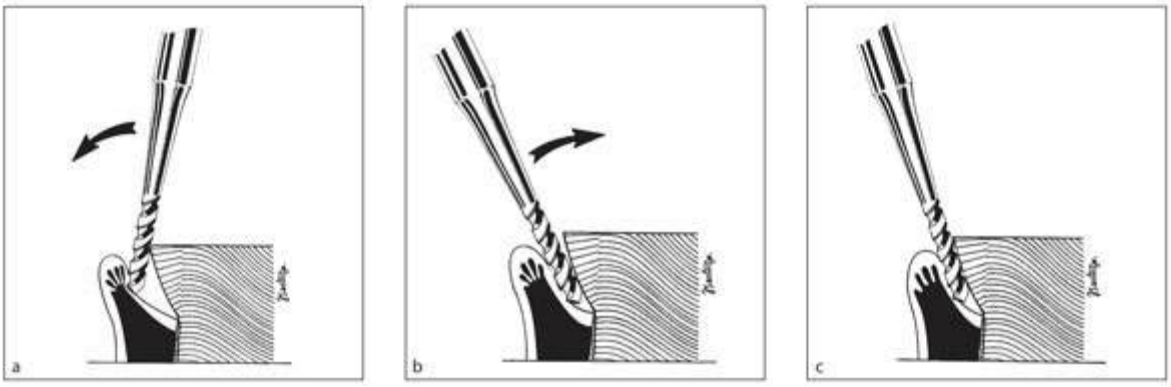


Fig 12-31*(a)* To align the pin channel drill with the side of the tooth when the external tooth surface is obscured, the drill is placed in the sulcus so that the drill is touching the preparation margin; the drill is then rotated *(arrow)*. *(b)* With that movement, the tip contacts the external surface and the portion of the drill that was in contact with the margin is rotated slightly away from it; then the length of the drill is rotated *(arrow)*. *(c)* With that rotation, the drill returns just to touch the margin. It is now aligned parallel to the external surface.

Pin channels should be initiated at least 0.5 mm from the DEJ if the nearby preparation margin is coronal to the CEJ; a 1.0-mm distance from the DEJ is preferable.[115] If the nearby margin is apical to the CEJ, there should be at least 1.0 mm of dentin between the channel and the external surface of the tooth.

The most common location for pins is at the line angles of the tooth because of the greater thickness of dentin between the external surface and the pulp and the decreased risk of perforation. The risk of perforation is especially increased in furcation areas. Figure 12-32 illustrates the preferred locations for pins.[122] Areas to be avoided in posterior teeth include proximal areas and tooth structure that lies over furcations or concavities in the root. Wherever the pin is to be located, the external surface of the tooth should be assessed and the pin channel drill aligned parallel to it (Fig 12-33).

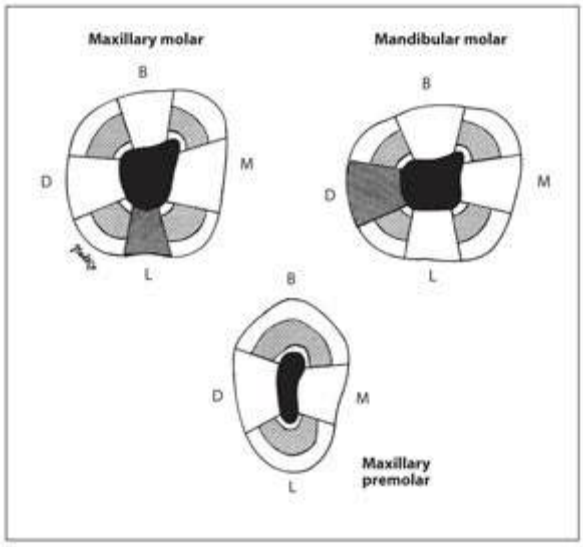


Fig 12-32 Preferred locations for pin placement. *(dotted areas)* The most preferred locations—at the line angles; *(white areas)* areas to avoid because of concavities, furcations, or thin dentin; *(lined areas on the molars)* areas where pins may be placed with added caution because the angulation of the root in relation to the crown is frequently severe.[122] B—buccal; M—mesial; L—ligual; D—distal.

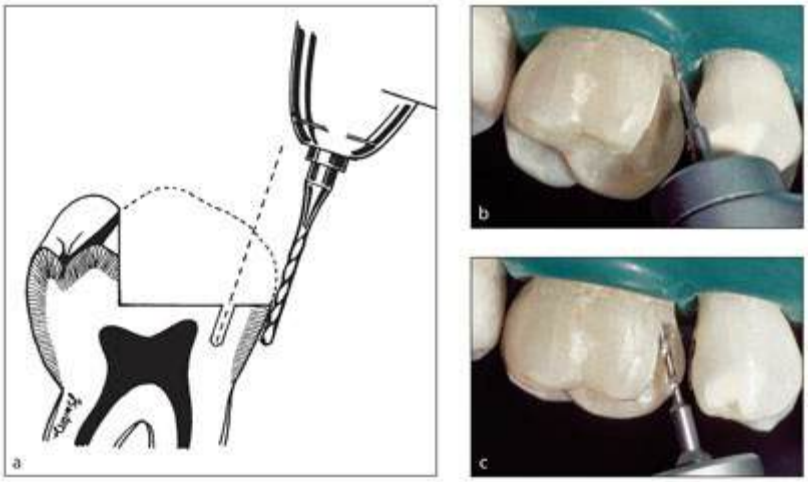


Fig 12-33*(a)* A non–depth-limiting pin channel drill is aligned parallel to the external surface. *(b)* A pin channel drill is aligned with the mesial surface of the maxillary molar before it is carried into the preparation. *(c)* The drill is used to cut the pin channel with the same alignment.

Pins should be located so that the channels enter the dentin at an approximately 90-degree angle to the prepared dentin surface. If a depth-limiting pin channel drill is used, the drill will not be able to achieve optimum pin channel depth if the surface of dentin adjacent to the entrance of the channel is at an angle to the drill. In addition, a pin should not be located immediately adjacent to a wall of the preparation; there should be access to condense amalgam around the full circumference of the pin. If a pin is located an optimal distance from the DEJ and a dentinal wall is adjacent to the

pin, a small “cove” may be cut in the dentin to provide adequate space for amalgam (Fig 12-34).

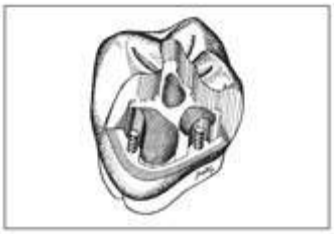


Fig 12-34 There should be adequate space around the full circumference of a pin for amal-gam. If a pin is located adjacent to a dentinal wall, a small cove may be cut in the wall to provide adequate space for condensation of amalgam.

To provide maximum cutting efficiency, the pin channel drill must be sharp so that it will be efficient at low speeds. A drill loses cutting efficiency with extended use and steam autoclaving.[123,124] Pin channel drills should be sterilized with dry heat, chemical vapor, or immersion in glutaraldehyde rather than in the steam autoclave. Drills should be discarded when a diminished cutting efficiency is sensed by the operator.[125,126] If preparation of a pin channel is difficult, it is likely that the flutes of the drill are obstructed by debris, the drill is dull, the handpiece is running in reverse, or the tip of the drill is in contact with enamel rather than dentin.

The correctly prepared pin channel will be slightly smaller in diameter than the pin; this size difference is called the *pin-to–pin channel mismatch*. The mismatch must be small to ensure that excessive stresses are not exerted on the dentin during insertion of the pin and that the pin will be retained by the dentin.

During preparation of a pin channel to a depth of 2 mm or more, it is advisable to withdraw the drill from the channel at least once to allow dentinal cuttings to be cleared from the flutes of the drill; this allows more efficient pin channel preparation and less heat generation. Care must be taken, however, to avoid overenlargement of the channel with multiple entries and withdrawals.

If an amalgam bonding material is used in addition to pins, it is applied after the pins have been inserted so that amalgam can be condensed immediately after placement of the bonding material.

Insertion. The insertion of self-threading pins may be accomplished in more than one way. They may be inserted by hand, using a small pin wrench (Figs 12-35a and 12-35b), or with a low-speed or finger-driven handpiece[127] (Figs 12-35c, 12-36a, and 12-36b). Placement by hand is preferred by some dentists because *(1)* it allows the operator to feel the insertion and to reverse the pin one-quarter turn once the tip

has contacted the end of the channel, thus avoiding excessive stresses in the dentin, and *(2)* stripping of the dentinal threads created by the self-threading pin is less likely. Insertion of pins with a low-speed handpiece is preferred by many because it is much more time efficient and the handpiece often allows easier access than inserting pins by hand. The thread design of some pins (such as the Link Plus series) provides for wider dentinal threads; dentinal stripping is less frequent when such designs are used.

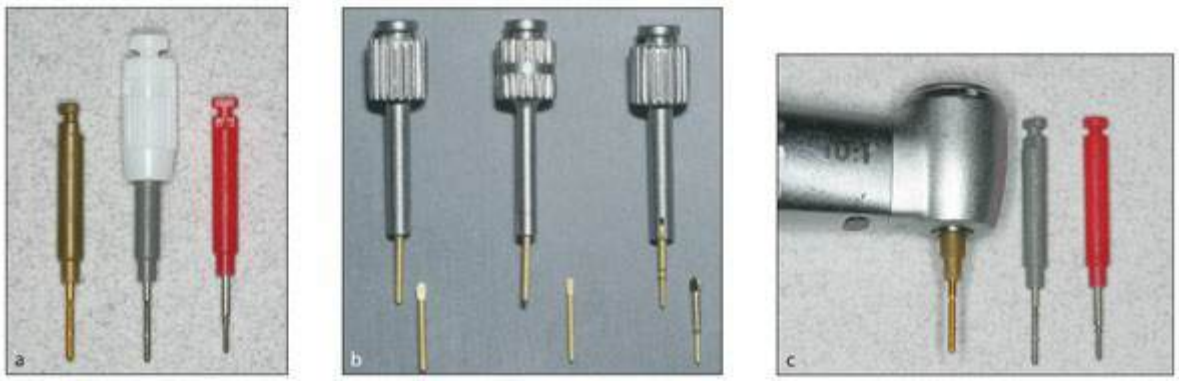


Fig 12-35 Pins and pin drivers (or wrenches) (TMS system). *(a) (left to right)* Link Plus Regular pin, Link Plus Minim pin, and Link Series Minikin pin. A plastic Universal Hand Driver (for Link Plus and Link Series pins) is shown on the Link Plus Minim pin. *(b)* Metal hand wrenches or drivers with individual TMS gold-plated stainless steel pins: *(left to right)* Regular, Minim, and Minikin. The hand driver with the band around the handle is for Minim pins; the nonbanded hand driver is for Regular and Minikin pins. *(c)* A motorized or hand-operated handpiece may be used to place Link Plus and Link Series pins. *(left to right)* Link Plus Regular gold-plated stainless steel pin in handpiece, Link Plus Minim titanium alloy pin, and Link Series Minikin titanium alloy pin.

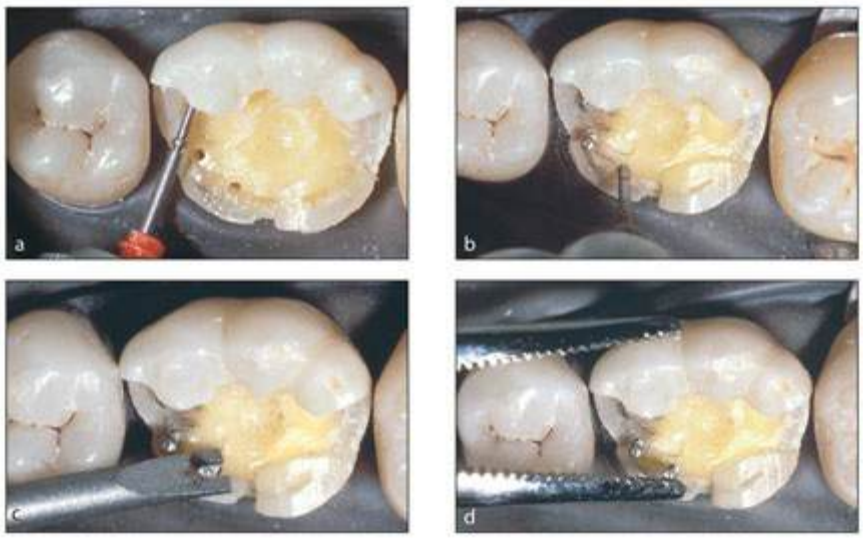


Fig 12-36*(a)* Insertion of a horizontal Link Series Minikin pin in a facial cusp of a mandibular molar. *(b)* A low-speed handpiece is used to insert a vertical Link Plus Minim pin. *(c)* To bend the pins, a Coltène/Whaledent pin bender can be used. *(d)* If a hemostat is used to bend the pin, the jaw of the hemostat should contact the pin at its tip, and very controlled pressure should be used in bending the pin.

Because the portion of a pin that will extend into the amalgam usually is only 2 mm long or less, the pin is aligned parallel to the external surface of the tooth, and the channel entrance is about 1 mm inside the DEJ, a pin rarely needs to be bent. The only reason for bending a pin is to keep it well within the bulk of the planned amalgam restoration. If bending should be needed, it may be accomplished with a small, fork-shaped pin bender (Fig 12-36c) or a small hemostat (Fig 12-36d) before the pin is shortened.

Shortening. The portion of a pin extending from the pin channel is often longer than desired, so it will need to be shortened after insertion. The pin may be efficiently and safely cut by gently brushing the pin with either a thin fissure bur or a diamond bur in a high-speed handpiece. The small segment of pin that is sectioned is a potential projectile that should be controlled with an instrument when possible. Both the bur and the diamond should be used with air or water coolant to prevent the pin, and therefore the surrounding tooth structure, from being overheated during the operation. If a bur is used, it must be sharp. If the bur approaches from an oblique angle, the clockwise rotation of the bur can cause counterclockwise rotation of the pin so that it is unscrewed. Therefore, if the pin cannot be approached perpendicularly by the bur (Fig 12-37a), it should be grasped with the pin wrench, cotton forceps, or a hemostat to stabilize it during the cutting process (Fig 12-37b), or, minimally, an instrument should be pressed against the pin during the process to dampen vibration from the bur, which tends to initiate the unscrewing. A long, narrow diamond is preferred by many operators for cutting pins, because it causes less vibration and is less likely to "catch" in the metal of the pin to initiate reverse rotation (Fig 12-37c).

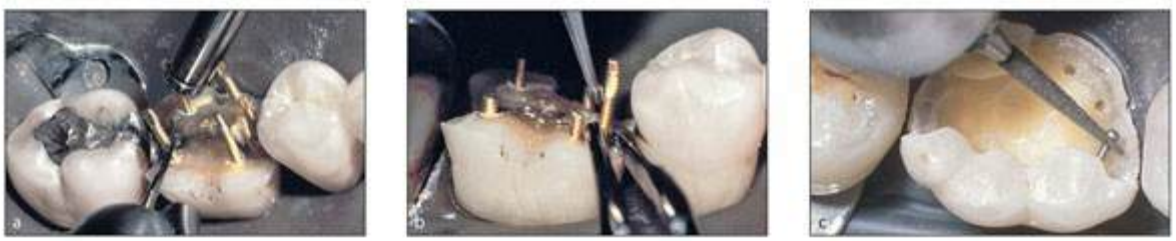


Fig 12-37 Shortening a pin. A light, brushing stroke and air or water coolant should be used while a pin is cut. *(a)* When a no. 169L tapered fissure bur is used to approach the pin at approximately 90 degrees, no stabilization is necessary. *(b)* When a no. 35 inverted cone bur is approaching the pin obliquely, the pin is stabilized to prevent it from being unscrewed from the pin channel. *(c)* This needle-shaped diamond is approaching the pin at approximately 90 degrees.

Horizontal pins

Studies[128,129] have demonstrated the efficacy of using pins oriented horizontally, that

is, inserted into the dentin of a vertical wall of a preparation (Fig 12-38). Burgess[128] found two horizontal self-threading pins (TMS Minim and Minikin) placed into a free-standing facial cusp of a maxillary premolar to be effective in reinforcing the cusp (see Fig 12-38a). Other investigators[129] have found that horizontal pins, used to cross-splint cusps of maxillary premolars, reinforce and strengthen the cusps (Fig 12-39).

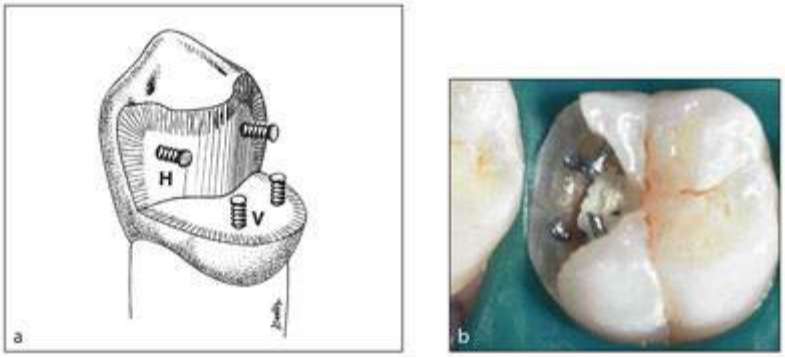


Fig 12-38*(a)* Horizontal pins (H) are used to attach the wall of a cusp to the amalgam restoration. Vertical pins (V) attach the restoration to the radicular portion of the tooth. *(b)* Horizontal pins can be used in conjunction with vertical pins. In this clinical situation, a proximal box had to be extended significantly facially and lingually to eliminate caries and unsupported enamel.



Fig 12-39 Horizontal pins are used to cross-splint the cusps of a maxillary premolar.

Adequate dentin must be present for horizontal pins to be employed. When the channels for horizontal pins are prepared, they should be initiated in dentin 0.5 to 1.0 mm from the DEJ. They should be directed approximately parallel to the adjacent DEJ (and external surface of the tooth). Because of their horizontal orientation, such pin channels, prepared only 1.5 to 2.0 mm deep, will often contact enamel. When the pin channel drill, in its progress through dentin, seems to stop its penetration short of reaching its depth-limiting shoulder, it is probably because it has reached enamel. Further deepening of the channel should not be attempted; even 1.0 mm of depth will impart some retention for a pin, and attempts to deepen the channel into enamel will result in an enlarged dentinal channel and increased potential for enamel fracture.

Horizontal pins should be positioned fairly near to the occlusal surface in the dentin of a vertical wall, 0.5 to 1.0 mm gingival to the occlusal DEJ, so that their mechanical advantage is enhanced for reinforcement of the cusp. A horizontal pin should be oriented so that it will not be near the anticipated surface of the amalgam and so that amalgam may be condensed around the entire circumference of the pin.

Perforation during pin channel preparation

Perforations during pin channel preparation should be avoided through careful design and placement of the channel. However, if a perforation does occur, it is important to determine what has been perforated, the external surface of the tooth or the pulp chamber. When the pulp chamber with a vital pulp has been perforated, the channel should be covered immediately with calcium hydroxide or mineral trioxide aggregate (MTA), and another pin channel should be placed in a new location. Alternatively, a different type of resistance feature should be used. A perforation of the external surface of the tooth may be more problematic. If the perforation is located above the epithelial attachment, the channel should be filled with amalgam. If the pin is inserted and the tip protruding on the external surface is cut even with the surface and polished, the pin will not totally obturate the perforation, and leakage will occur.[130] If the perforation occurs below the epithelial attachment, the channel may be obturated with MTA or with amalgam.

Nonpin mechanical resistance and retention features

Birtcil and Venton[131] suggested that more attention be directed toward using the available tooth structure to provide retention and resistance form in complex amalgam restorations. They recommended parallelism in all walls of the preparation, proximal box form, retention grooves in the proximal line angles, box form in buccal and lingual groove areas of molars, dovetails, rectangular boxes in areas other than proximal surfaces, and reduction of undermined cusps for coverage with amalgam. In recent years, several additional nonpin resistance and retention methods have been described and investigated. These include the circumferential slot and the amalgapin, as well as adhesive bonding (described in the Amalgam Bonding section and in chapter 6).

Circumferential slots. Outhwaite et al,[132] who introduced the circumferential slot prepared with a no. 33½ inverted cone bur (Fig 12-40), compared it with four pins (TMS Minim) in an in vitro study and found no significant differences between the

resistance provided by the two techniques. They also reported that the pin restorations had a greater tendency to slip on their bases before failure, whereas slippage did not occur with circumferential slots. However, slot-retained restorations are more sensitive to displacement during matrix removal than are pin-retained restorations. In the design of preparation resistance form, *segmental slots* (short segments of circumferential slots) are used effectively. Some operators have called these segmental slots "cleats."

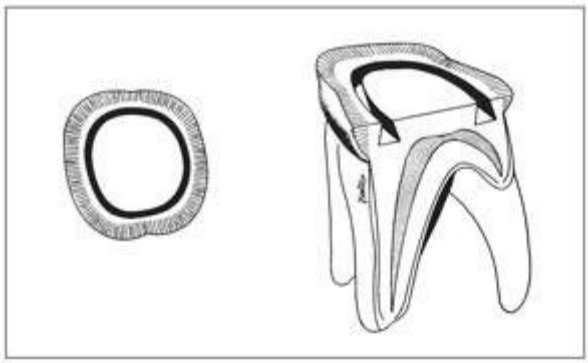


Fig 12-40 A circumferential slot is prepared with a small, inverted cone bur, such as a no. 33½.

Amalgapins. Seng et al[133] tested circular chambers that they cut vertically into dentin to provide resistance and retention form for the restoration; they called these features *amalgam inserts*. Preparations for the inserts were made with a no. 35 inverted cone bur and were approximately ¼ mm in diameter and depth. In their study, amalgam inserts provided resistance to displacement similar to that provided by self-threading pins. Shavell[134] described a variation of the amalgam insert, which he termed the *amalgapin* (Fig 12-41). The amalgapin channel described by Shavell was prepared with a no. 1157 or 1156 bur and had a depth of 3.0 mm. Laboratory studies of the amalgapin[135,136] have demonstrated that the resistance to displacement provided by amalgapins is similar to that provided by pins. It has been demonstrated[136] that a depth of 1.5 to 2.0 mm is adequate for amalgapins and that an amalgapin with a diameter of 0.8 mm provides resistance similar to that of an amalgapin with a diameter of 1.0 mm. In addition to the burs advocated by Shavell (nos. 1156 and 1157), others with similar diameters (such as the nos. 330 and 56) also function well in creating amalgapin channels.[136]

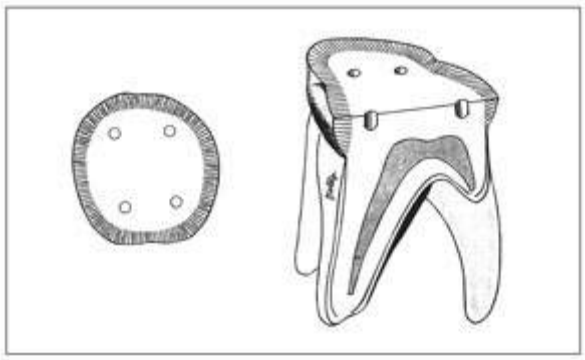


Fig 12-41 Amalgapin channels are prepared with a diameter of 0.8 to 1.0 mm, such as that of a no. 330 bur, to a depth of approximately 1.5 mm.

Efficacy of resistance and retention methods

For the most part, the resistance form provided by various resistance and retention methods has been tested in flattened molars, as described by Buikema et al,[121] with 4.0-mm-high restorations retained by given macromechanical resistance features or amalgam bonding. These teeth were mounted at a 45-degree angle and were loaded in compression; the mean loads at the time of failure were calculated. Although this method of testing is not as enlightening as long-term clinical tests, it probably provides a good indicator of how well a resistance feature will perform in a clinical situation. It has been shown, however, that if these standard resistance-test restorations are loaded at a 90-degree angle instead of a 45-degree angle, the stainless steel pins provide significantly more resistance than amalgam inserts.[137,138] Few forces in the mouth are directed at a 90-degree angle to the long axis of the tooth, however, and few restorations are placed on preparations that are totally flat, without any walls or irregularities in the dentin.

One of the most telling studies pertaining to resistance form for complex amalgam restorations was reported by Plasmans et al.[139] This group created preparations for complex amalgam restorations that combined the use of boxes, shelves, and amalgapins as resistance and retention features. They then loaded specimens at 45 degrees, as in most previous studies of resistance form, but they loaded half of the restorations from one side and half from the other. Their finding was, generally, that more load was required to cause failure of a restoration when the resistance and retention features (walls, boxes, and amalgapin channels) that opposed the direction of the load were increased.

It is important to distribute mechanical resistance features into all areas of the preparation and not to cluster them in any one area.[138–140] Figures 12-42a to 12-42d show a restoration that originally replaced two missing cusps of a mandibular molar.

The probable cause of failure was that the resistance features (pins) were clustered in the lingual aspect of the cavity preparation. In function, there was nothing to attach the facial aspect of the restoration to the tooth. If two of the four vertical pins had been placed in the facial portion of the preparation (Fig 12-42e), or if two horizontal pins had been placed in the facial cusps (Figs 12-42f and 12-42g), the restoration would, in all likelihood, have had adequate resistance to withstand its load in function.

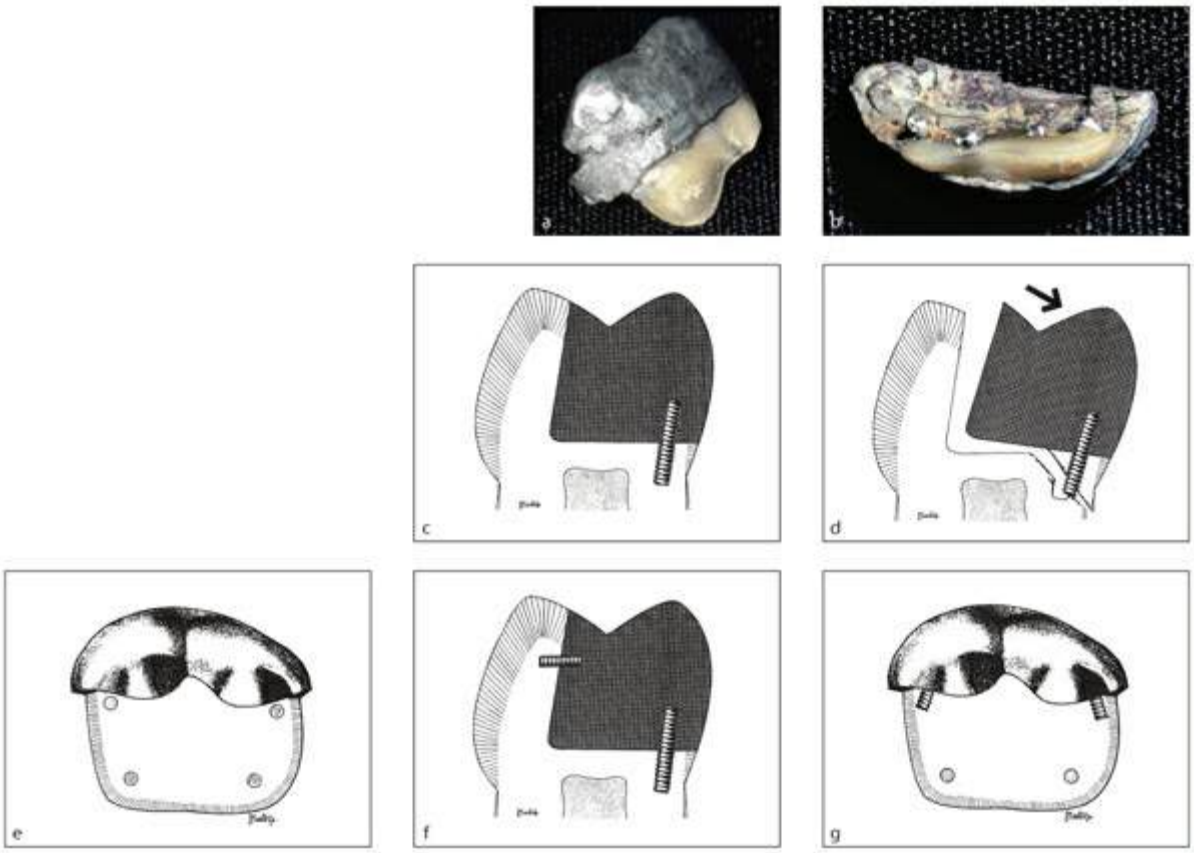


Fig 12-42*(a and b)* Failed complex amalgam restoration that replaced the lingual cusps of a mandibular molar. Note the fin of cervical tooth structure that fractured. *(c)* The dentin lingual to the pin was the only tooth structure that was opposing a lingually directed load on the restoration. *(d)* A load pushing the restoration lingually caused failure. Failure can be attributed to a lack of distribution of resistance and retention features. *(e)* Alternatively, had two of the four pins been placed vertically in the facial aspect of the preparation, this could have reduced the likelihood of failure. *(f and g)* Had two of the four pins been located horizontally, failure would likely have been averted.

When the technical requirements for placement of vertical pins can be met, vertical pins provide excellent retention and resistance form. However, risks are involved with pin placement: crazing of tooth structure, perforation into the pulp or periodontium, and weakening of the amalgam restoration over pins.[141] Additionally, the use of both vertical and horizontal pins may be limited by inadequate access; in

these cases, alternative resistance and retention methods should be employed. When a cusp has been reduced and increased resistance form is needed, adhesive bonding, an amalgapin, or a segment of a circumferential slot may be indicated.

Amalgam Bonding

The use of adhesive resins to increase the retention, resistance, and marginal seal of amalgam restorations has gained a strong foothold in restorative dentistry. There is now more than adequate evidence that properly bonded amalgam restorations will be as successful as pin-retained amalgam restorations. Several clinical studies have demonstrated success after several years of service.[107,142–145] In one clinical study of complex amalgam restorations,[142] all bonded and pin-retained restorations were classified as successful at 2 years. Six-year results were reported for another clinical study[107] in which 32 bonded (Amalgambond Plus with HPA powder, Parkell) and 28 pin-retained (TMS pins) amalgam restorations were compared; each restoration replaced at least one cusp, and, in some of the restorations, the only sources of retention were the pins or the adhesive bonding (see example in Fig 12-74). After 5 years, seven of the pin-retained and two of the bonded restorations had failed (N = 46). Failures of bonded restorations involved loss of another cusp. None of the bonded amalgam restorations had debonded. At 6 years, the results continued to demonstrate that bonded amalgam restorations performed as well as pin-retained amalgam restorations, with failure of one from each category in the 6th year (N = 27). The number and length of clinical studies[146] is now sufficient to indicate that bonded amalgam restorations can perform as well as restorations using mechanical resistance and retention features when the bonding technique is accomplished properly.

Pins and bonding can have an additive effect in which the resistance to displacement is superior to either technique alone.[147–150] In one in vitro study,[147] the mean resistance to dislodgment provided by a filled amalgam bonding material combined with self-threading pins was approximately equal to the sum of the mean resistance produced by pins alone and the mean resistance produced by bonding alone.

For use of bonding resins in amalgam restorations, research suggests the improved efficacy of filled resins compared with unfilled or minimally filled resins.[150–153] The method of incorporation of the filler varies from manufacturer to manufacturer. One system (Amalgambond Plus) uses very fine methyl methacrylate powder, added to

the liquid resin, as the filler.[154] Another system (All-Bond 2 and Liner F, Bisco) uses a filled flowable resin composite liner to provide the filled resin. With both systems, the amalgam is condensed into the filled resin while the resin is in a viscous liquid form. Microscopic "fingers" of resin are incorporated into the amalgam at the interface. When hardened, these provide the attachment of amalgam to resin. Because light cannot penetrate to the resin underlying amalgam restorations, it is important to use a self-curing or chemically activated bonding resin. The bonding resin of an amalgam bonding system is supplied in two parts that are to be mixed. The attachment of resin to tooth structure when amalgam bonding systems are used is accomplished as with other dental bonding systems, as described in chapter 9.

There is ample short-term evidence of decreased leakage of fluids when an amalgam bonding system is used compared with noncoated or varnish-coated amalgam cavity walls.[155–160] Many dentists have reported that amalgam bonding systems have reduced the incidence of postoperative tooth sensitivity, but most research has not supported these observations. Clinical studies[161,162] comparing postoperative sensitivity in bonded and nonbonded restorations have reported no significant difference. One study[163] involving 40 teeth with symptoms of cracked tooth syndrome prior to restoration found no difference in thermal sensitivity between pin-retained and bonded restorations at 2 weeks but found less thermal sensitivity in teeth with bonded restorations at 3 and 12 months.

For reinforcing cusps in posterior teeth weakened by significant pathosis or large restorations, there is in vitro evidence that amalgam bonding can result in some strengthening of teeth.[164–167] However, much of this effect may be lost with time.[168–170] At present, it is still advisable to reduce severely weakened cusps for replacement and protection with a strong restorative material.[141] The use of amalgam for this purpose is described in this chapter. Other types of occlusal-coverage restorations are described in chapters 19 and 20.

Evidence from clinical and laboratory studies supports the contributions to resistance to displacement that come from the various resistance features and methods (cavity form, threaded pins, slots, grooves, amalgapins, and filled adhesives). The resistance mechanisms used in any restoration should be selected based on the functional load to which the restoration is expected to be exposed. Amalgam bonding probably may be used as the sole means of retaining an amalgam restoration, but it will be more effective for supplementing mechanical resistance features in large, complex amalgam restorations, especially those replacing cusps.

Most preparations have some box form and surface irregularities that, when combined with resin bonding, can provide long-term retention of the restoration. A filled amalgam bonding system should be used. Amalgam bonding should be used when an improved initial seal is needed, such as after a direct or indirect pulp capping procedure in the tooth being restored. Adequate moisture control and meticulous attention to product instructions are necessary to duplicate the success that has been demonstrated in clinical trials.

Matrices

To confine the amalgam and allow adequate pressure for optimum condensation, the preparation must be boxlike, that is, with confining walls and floors. A Class 1 occlusal preparation provides these by virtue of its location and definition. Whenever an amalgam preparation extends from one surface of the tooth to another, some form of matrix is needed to confine the amalgam for condensation. If a matrix is not used, condensation forces will tend to push the amalgam out of the preparation rather than condensing the amalgam.

The simplest and fastest method of adequately providing a matrix should be used. Occasionally, as with the occlusolingual restoration of the maxillary molar, the blade of a hand instrument, placed on the lingual surface of the molar and held in place during condensation, may be adequate. In most cases, a matrix system, such as a Tofflemire matrix, is indicated. Rarely, a customized matrix will be necessary for a particular situation. A matrix system is needed in almost every case for multisurface Class 2 amalgam restorations.

There are myriad types of matrix systems; most involve thin pieces of stainless steel that are contoured and placed adjacent to the proximal portions of Class 2 preparations. The purpose of the matrix is to:

- Confine the amalgam so that adequate condensation forces can be applied.
- Allow re-establishment of contact with the adjacent tooth.
- Restrict extrusion of the amalgam and the formation of an overhang at a hidden margin, such as the proximal gingival margin.
- Provide for adequate physiologic contour for the proximal surface of the restoration.
- Impart an acceptable surface texture to the proximal surface, especially in the area of the contact that cannot be carved and burnished.

Tofflemire matrix

Probably the most commonly used type of matrix system in the United States is the Tofflemire system. The system consists of a matrix band and retainer (Fig 12-43). The Tofflemire retainer consists of four parts (see Fig 12-43b).

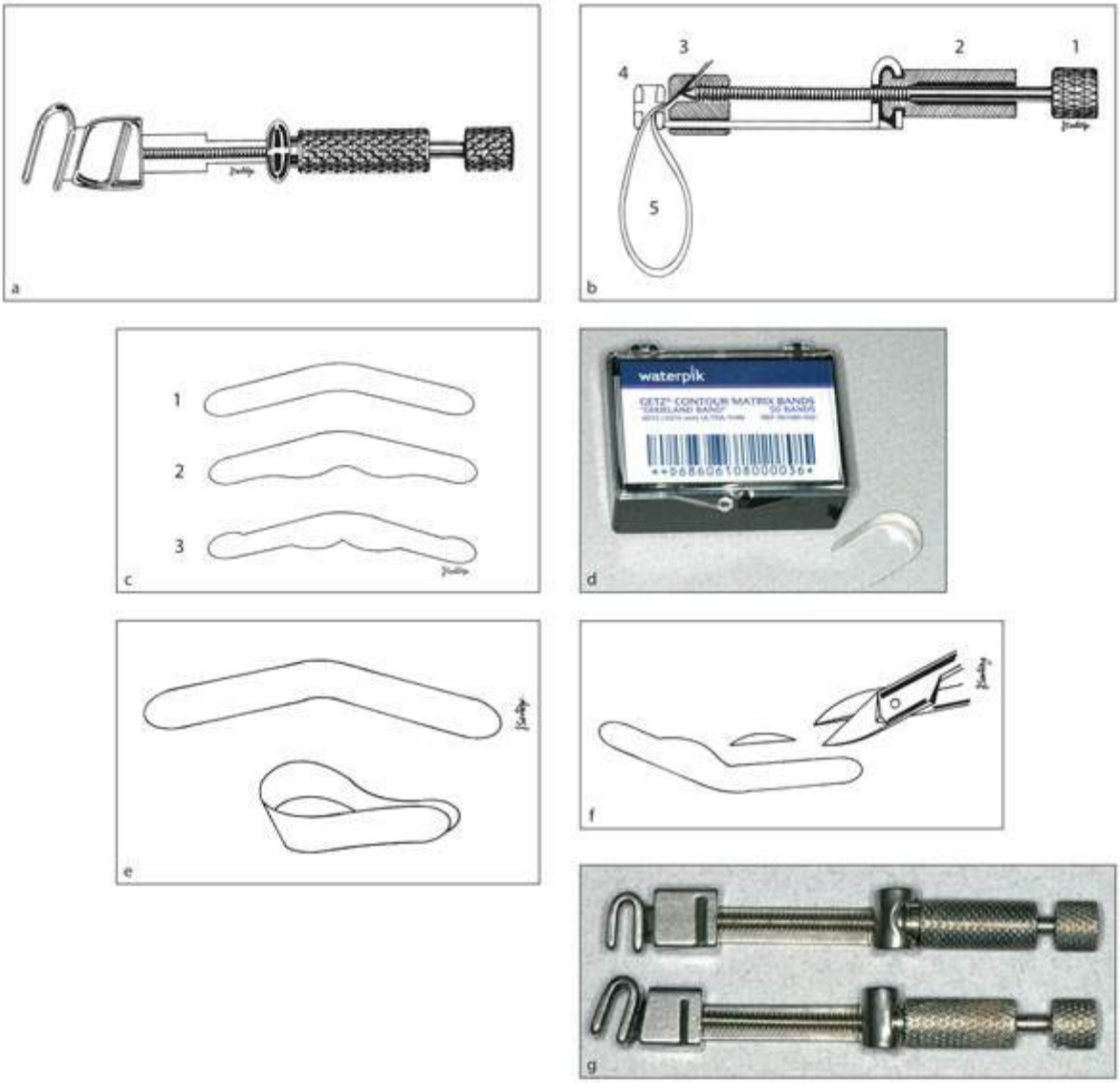


Fig 12-43 Tofflemire matrix system. *(a)* Matrix retainer. *(b)* Parts of the assembly: set screw (1), rotating spindle (2), slide (3), head (4), band (5). *(c)* Three common shapes of Tofflemire matrix flat bands (no. 1 is also called a *Universal band*; no. 2 and no. 3 are also called *MOD bands*). *(d)* Precontoured Dixieland Bands. *(e)* The matrix band is folded for insertion into the retainer. *(f)* One of the projections of a no. 2 or 3 band may be cut off if there is only one deep proximal area of the preparation. *(g)* Two types of Tofflemire retainer: *(top)* straight; *(bottom)* contra-angled.

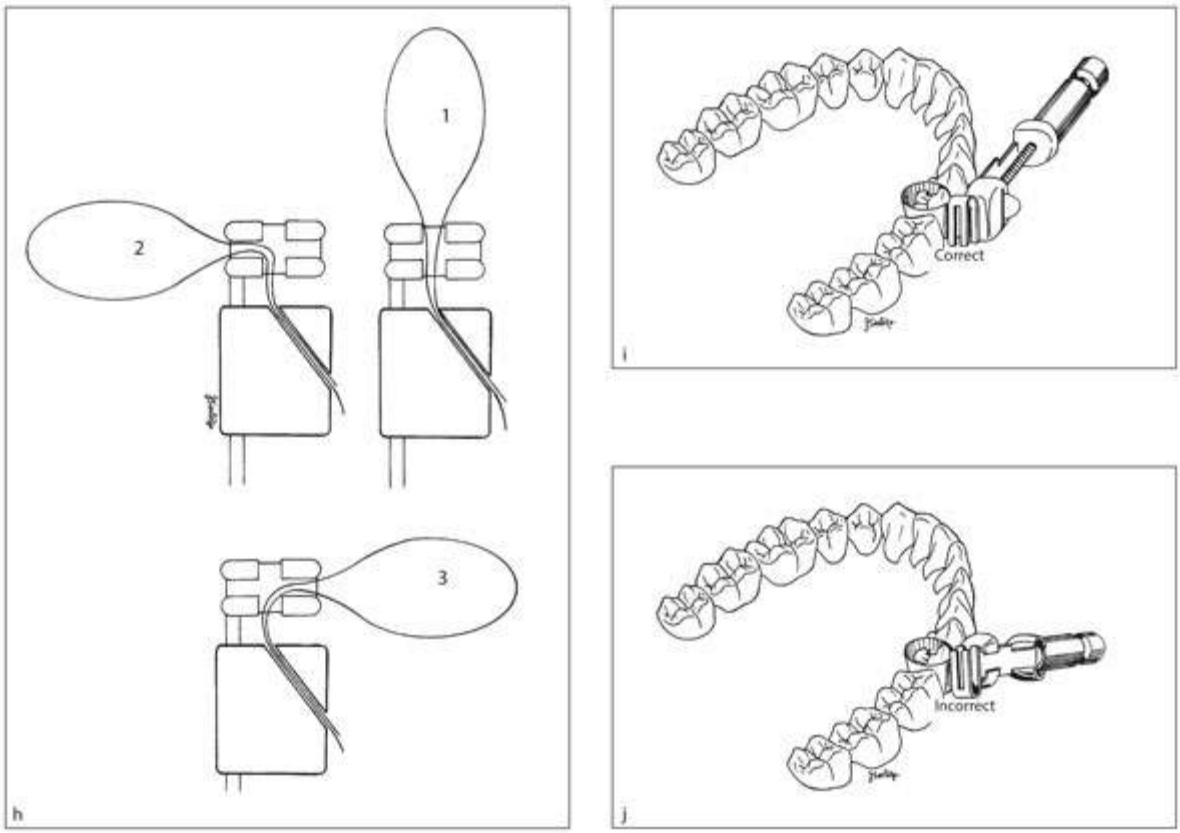


Fig 12-43*(cont) (h)* The loop of the band may extend from the head of the retainer in one of three directions: straight (1), left (2), right (3). *(i)* The matrix must be assembled with the slots in the head directed gingivally. *(j)* The slots in the head of the matrix should not be directed occlusally.

Parts

Head. This is the part that has the open side. In the U-shaped head, there are two slots in the open side. These slots are used to position the matrix band. The open side of the head should be held facing upward while the band is installed. The open side of the head faces gingivally when the band is placed around the tooth. There are two types of Tofflemire retainers based on the angulation of the head (see Fig 12-43g).

Slide. This element has a diagonal slot. The round ends of the band, when installed, extend at least 1 to 2 mm beyond the slot in the side of the slide. The amount of the band extending beyond the slot in the side of the slide will depend on the size of the tooth being treated. More of the band should extend from the slide for premolars than for molars. The slide is positioned near the head for installation of the band in the retainer and for placement of the band around the tooth.

Rotating spindle. This is used to adjust the distance between the slide and the head.

The retainer is held with the thumb and forefinger of one hand (contacting both the head and slide) while the rotating spindle is turned with the other hand, clockwise and counterclockwise, to advance and retract the slide. This movement adjusts the size of the loop of the matrix band.

Set screw. The threaded shaft of the set screw locks and unlocks the matrix band in the slide.

Types

The types of Tofflemire bands include flat bands of multiple shapes, precontoured bands, and bands with and without memory (dead-soft metal).

Flat bands. Bands for the Tofflemire system come in several shapes (see Fig 12-43c) and in three thicknesses: 0.0010, 0.0015, and 0.0020 inches. The thicker band is stiffer to resist deformation during condensation; the thinner bands are often used to help ensure a tight contact in Class 2 restorations. Any of the three thicknesses can be used to achieve excellent results, and selection is primarily a matter of operator preference.

By far the most frequently used shape is the no. 1, or Universal, band. The no. 2 (so-called MOD) band has two extensions projecting at its gingival edge to allow matrix application in teeth with very deep gingival margins in the proximal aspects of the tooth. In most cases, there will be only one deep area, so one of the extensions is usually cut off with scissors (see Fig 12-43f). The no. 3 band also has projections for deeper gingival margins, but the band is narrower than the no. 2 band. The no. 3 band is ordinarily considered suitable for premolars and the no. 2 band for molars, but the size that best suits the situation should be used.

Because these bands are flat, they should be contoured so that they will impart physiologic contours to the restorations. A flat band may be contoured before it is placed in the retainer. The band is laid on a paper pad or other compressible surface, and the area to be contoured is heavily rubbed with an ovoid burnisher, a beavertail burnisher, the convex back of the blade of a spoon excavator, or a convex side of the cotton forceps. A band may also be contoured after it has been applied to the tooth. The area to be contoured is rubbed with the back of the blade of the spoon excavator or with some other thin, convex instrument (Fig 12-44a). Contact with the adjacent tooth should be more than a pinpoint touch (Fig 12-44b).

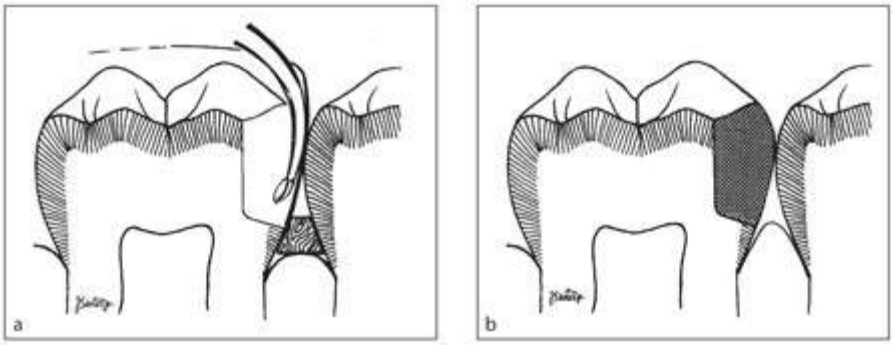


Fig 12-44*(a)* The convex side of a spoon excavator is used to impart a convex contour to the matrix band. *(b)* This will help to achieve a contact area, as opposed to a pinpoint contact, with the adjacent tooth.

Precontoured bands. Precontoured Tofflemire matrix bands are also available. One such band is the Getz Contour Matrix Band or Dixieland Band (Waterpik Technologies) developed by Dr Wilmer Eames (see Fig 12-43d). When these bands are removed from interproximal contacts, the contour must be considered, and the band must be rotated in such a way that the trailing edge does not break or alter the shape of the marginal ridge as the band is being removed.

Placement

Assembly. When the matrix retainer and band are assembled, the two ends of the matrix band must be even as they protrude from the diagonal slot of the slide. The loop can extend from the retainer in three different ways: straight, to the left, or to the right (see Fig 12-43h). The straight assembly is for restorations near the front of the mouth where the rubber dam–covered cheek will not get in the way if the retainer protrudes perpendicularly from the line of teeth. The right and left assemblies allow the retainer to be aligned parallel or tangent to the line of teeth in more posterior areas. The band should be placed in the retainer so that the loop extends from the appropriate side of the retainer and the set screw knob is directed toward the front of the mouth. Because of the shape of the Tofflemire matrix band, when it is placed in the retainer, one opening of the loop has a greater diameter than the other (see Fig 12-43e). In other words, the loop will be shaped somewhat like a funnel. The wider opening is oriented toward the occlusal aspect. The short knob of the set screw is tightened so that the matrix band is held securely.

Application. The matrix is applied to the tooth to be restored. The matrix band will slide easily through the interproximal contact area when the preparation has opened, at least partially, the contact. It will often slide through an intact contact as well (eg,

the mesial contact when there is a disto-occlusal preparation). If it will not slide through the intact contact, a bladed plastic filling instrument (such as the no. 1-2, shown in Fig 12-45) may be used to open the contact slightly to allow the band through. For this slight tooth separation, the blade of the instrument is placed in the gingival embrasure (gingival to the contact), moved occlusally until it is stopped by the contact, and rotated (torqued) very slightly. At the same time, the matrix band is slipped through the contact. When the matrix is around the tooth, it should be tightened snugly, but not too tightly, because a very tight matrix will deform the tooth.[171–173]



Fig 12-45 The blade of a plastic filling instrument has been placed into the gingival embrasure and is being slightly rotated (torqued) to provide enough separation to allow the matrix band to slip through the contact.

Wedging. Wooden wedges may be placed from either the facial or the lingual aspect. A wedge should usually be inserted from the side with the widest embrasure. For example, between the first and second premolars, the largest embrasure is usually the lingual embrasure. The wedge should be inserted tightly to enable development of an adequate contact despite the thickness of the matrix material.

Wedges are available in a variety of shapes and sizes. Figure 12-46a shows Wizard wedges (Waterpik Technologies), which are triangular and available in four sizes. Figures 12-46b and 12-46c show Sycamore wedges (Premier), which are shaped to aid the establishment of physiologic proximal contours. The Premier wedges, with seven color-coded sizes, are recommended for amalgam restorations.



Fig 12-46 Wooden interproximal wedges. *(a)* Wizard wedges have a triangular shape. *(b)* Premier Sycamore wedges are shaped to impart a more physiologic contour to the matrix. There is a larger selection of sizes, and they are color coded for easy selection. *(c)* Note the anatomical

shape of the Premier Sycamore wedges. The snow-sled point helps to prevent catching of the rubber dam during insertion.

The matrix band must extend gingival to the gingival margin of the proximal box of a Class 2 restoration, and the wedge must be positioned so that its base is also gingival to the gingival margin. If the wedge cannot be placed so that its base is gingival to the preparation margin, a concavity will be created in the matrix just occlusal to the gingival margin, and this concavity will be transferred to the amalgam. Occasionally, the gingival papilla will need to be surgically reflected from the interproximal area to allow the wedge to be positioned apical to the gingival margin. Another option is to use a rigid bladed instrument to hold the matrix against the gingival margin during condensation. Custom wedges may be made for special situations.

Contouring. The band should be burnished and contoured to impart the desired proximal contours to the restoration. This can be accomplished with the back (convex side) of a spoon excavator (see Fig 12-44a). The wedged matrix should be in solid contact with the adjacent tooth in the desired contact area. It should be possible to feel the convexity of the proximal surface of the adjacent tooth with an instrument through the matrix as the matrix is burnished.

Removal

In a multiple surface restoration, amalgam is condensed in the preparation after matrix placement. Amalgam condensation is discussed in the next section.

When amalgam placement using a Tofflemire matrix has been completed and preliminary carving of the occlusal aspect accomplished, first the slide and then the set screw of the matrix retainer is loosened. A finger or thumb is placed on the loop of the matrix band to keep it in place on the tooth, and the retainer is pulled occlusally to remove it. The distal end of the matrix band is grasped and pulled occlusally and lingually (if the free ends are on the facial aspect) and out of the distal contact of the tooth. The mesial end is then grasped and pulled facially and occlusally until the band is out of the contact. The matrix band can be grasped with fingers, cotton forceps, or a hemostat. There are a few techniques that may help the dentist remove the Tofflemire matrix without breaking the marginal ridge:

- As the matrix edge is coming out of the contact, the matrix can be tipped so that the edge will not “flip” the newly carved marginal ridge and break it.

- A condenser can be held against the marginal ridge to support it and prevent it from breaking as the matrix is removed.
- The movement of the band should be primarily to the facial or lingual aspect as the band slips occlusally out of the contact.

The matrix band should be used only once and then discarded.

Other matrix systems

Many matrix systems other than Tofflemire matrices are available (Figs 12-47 to 12-51). Each has its own advantages and disadvantages. In addition, stainless steel matrix material may be spot-welded to provide a custom matrix for any situation. One commercial system (Denovo) has prewelded bands in various sizes (see Fig 12-47). To remove a spot-welded matrix from a tooth after the restoration is placed, a small bur in a high-speed handpiece is used to cut through the welds and allow the two ends of the matrix to separate. The absence of a matrix retainer in the Denovo system is a distinct advantage.



Fig 12-47*(a and b)* Denovo matrix system.

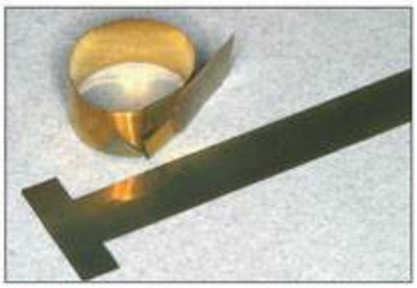


Fig 12-48 T-band matrix.

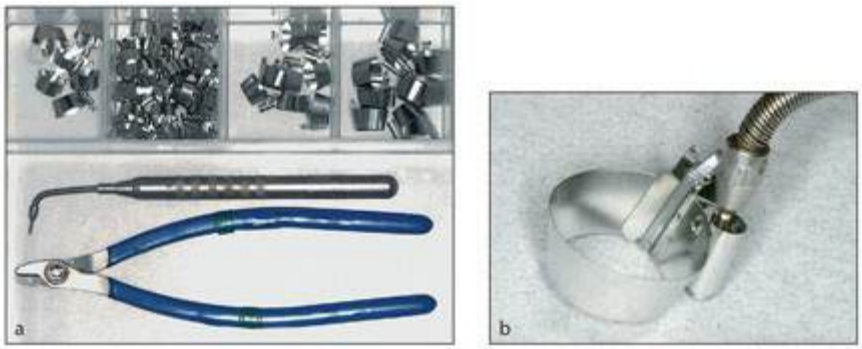


Fig 12-49*(a and b)* AutoMatrix system (Dentsply). Note the cable-drive wrench for adjusting the size of the loop.



Fig 12-50*(a and b)* Palodent matrix system (Dentsply/Caulk). Note the metal ring for holding the ends of the matrix snugly against the facial and lingual enamel. It will also provide some tooth separation.



Fig 12-51*(a and b)* Ring systems made for use in placing resin composite restorations (such as these Composi-Tight Soft-Face 3D rings, Garrison) provide excellent matrices for amalgam as well.

T-bands have long been used in dentistry and provide a very simple and inexpensive matrix system (see Fig 12-48). The AutoMatrix system has a built-in matrix retainer that is much smaller than the Tofflemire matrix retainer, which is an advantage (see Fig 12-49). The Palodent matrix and some other matrix systems provide small, precontoured matrices that are placed, wedged, and held in place by a flexible ring. A major advantage of this system is that, for a restoration involving only one proximal surface, there is no need for the matrix to be placed in the other

contact (see Fig 12-50). If two proximal surfaces are to be restored, two small matrices, two wedges, and two rings may be used (Fig 12-51).

Reinforcing matrices with modeling compound

Among the desirable qualities of a matrix are adequate rigidity and the ability to maintain the shape established by the operator to impart the desired contour to the restoration. When a Class 2 preparation has only proximal boxes that are adjacent to other teeth, and when the preparation does not to any significant degree extend to facial and lingual surfaces, the stainless steel matrix is usually well supported by the adjacent tooth or teeth. In these cases, no reinforcement is necessary. In larger restorations that involve surfaces not supported by adjacent teeth, it is often desirable to reinforce or support the matrix in some way in these areas to maintain the rigidity and shape of the matrix.

Occasionally, a single unsupported area of a matrix may be reinforced during condensation by the operator, who places a finger or holds an instrument against the matrix in a facial or lingual area. For large unsupported areas, however, modeling compound may be used (Fig 12-52).



Fig 12-52 Modeling compound can be used to support a matrix. *(a and b)* The compound stick is heated over an alcohol flame, then removed from the flame to allow warmth to diffuse to the core of the stick. *(c)* When the warmed tip of the compound stick begins to droop, softness is uniform throughout, and the compound is ready for use. *(d)* A finger is dampened in water to prevent the

glove from sticking to the softened compound. *(e)* The compound has been pressed into place. It will be cooled with air to reharden it. *(f)* Compound has been broken into smaller pieces and inserted into a plastic syringe, which in turn is placed into a warm water bath. *(g)* Once the compound has softened, it can be easily ejected. Note that the tip has been shortened to provide a wider lumen so that the compound extrudes easily. *(h)* The matrix may be recontoured after application of the compound. A warmed instrument is used to soften the compound and reshape the matrix. *(i)* Any compound extending past the edge of the matrix should be trimmed to prevent chipping during amalgam condensation.

There are various ways of applying compound to support a metal matrix. Probably the simplest is to employ a stick of compound (see Fig 12-52a). Approximately 1 inch of one end of the compound stick is heated over an alcohol burner. The stick is moved back and forth, while being rotated, over the tip of the flame. After 5 to 10 seconds, the stick is removed from the flame (see Fig 12-52b) and held for a few seconds until the heat has diffused through the radius of the stick to its center, as indicated by its starting to droop or sag (see Fig 12-52c). At that point, the 1-inch end is soft enough to carry to the matrix and press into place with a dampened, gloved finger (see Figs 12-52d and 12-52e). If adhesion of the compound to the matrix and adjacent teeth is desired, the softened end of the stick should be passed through the flame again just before it is carried to the mouth; this will provide a tacky surface that will impart some adhesion. An alternative way to warm and deliver compound is to break up a stick of compound into smaller pieces and insert them into a plastic syringe. This syringe can then be placed into warm water so that the compound softens (see Fig 12-52f). Once softened, it is then possible to extrude the warmed composite from the syringe (see Fig 12-52g); it may be necessary to cut off a portion of the syringe tip to ensure an adequate lumen size to allow the composite to flow easily. This makes it straightforward to inject compound into the areas needed to stabilize the matrix or rubber dam clamp. The compound can be adapted with a dampened, gloved finger as described earlier.

After the compound is pressed into place (see Fig 12-52e), it is cooled and hardened with air from a three-way syringe. The matrix may be recontoured after compound application. A warmed instrument may be used inside the matrix to soften the compound and exert pressure on the matrix to give it the shape that will allow the restoration contours and shape to be similar to the original shape of the tooth (see Fig 12-52h). Again, the compound should be cooled with air after reshaping with a warmed instrument. If modeling compound extends occlusal to the occlusal edge of the matrix band, it should be trimmed back with a sharp instrument (see Fig 12-52i); otherwise pieces of compound could chip off during amalgam condensation and contaminate the amalgam.

If condensation forces dislodge the compound, matrix reinforcement will be lost; steps should be taken, therefore, to ensure that the compound does not dislodge. While it is soft, a portion of it may be pushed onto the cusps of an adjacent tooth to provide retention. The compound can usually be pried away from the adjacent teeth and matrix with an instrument such as a Hollenback carver or enamel hatchet. After the compound is removed, the matrix may be removed as previously described.

Matrices for bonded amalgam restorations

If a resin bonding system is used to coat the walls of the preparation, care should be taken to prevent or minimize resin application to the matrix. If resin is applied to the matrix, it may cause the matrix to stick to the amalgam. This sticking can lead to fracture of the amalgam during removal of the matrix. Attachment of the matrix to the amalgam is most significant when amalgam bonding materials are used. Because amalgam must be inserted immediately after placement of the adhesive, the bonding material cannot be placed before matrix application. A very small applicator should be used to apply the resin to the preparation walls so that it may be kept away from the matrix. It is advisable to try to stop the resin application approximately 1 mm short of the cavosurface margins that are adjacent to the matrix. Unless the set of the material is too advanced by the time the amalgam is placed, it will be pushed to the margin in a thin coat as the amalgam is condensed.

Matrices that resist adhesion to the bonding materials are available, but if a band that is specifically designed to prevent adhesion to the matrix is not used, the application of a very thin coat of wax with a wax pencil or crayon or with a piece of inlay wax or boxing wax may be helpful. The wax is rubbed onto the inner surface of the matrix band, and excess is rubbed off with a gloved finger.

Placement of Amalgam

The technique for amalgam placement is basically the same regardless of the type or classification of the preparation. Amalgam is mixed (triturated), carried to the cavity preparation, and condensed into the preparation so that voids are eliminated and all areas of the preparation are filled. The amalgam is then carved to reproduce the portion of the tooth that is missing.

Spherical alloys produce an amalgam that requires a lower mercury-alloy ratio and less condensation force. However, the direction of the condensing force is extremely important for spherical amalgams. They do not adapt to the cavity walls as

well as lathe-cut or admixture amalgams.[174] Spherical amalgams are said to be less condensable, and lateral condensation is even more important when spherical amalgams are used than when admixture amalgams are used. It is also somewhat more difficult to obtain good interproximal contacts in Class 2 amalgam restorations with spherical amalgams than with lathe-cut amalgams or admixtures. It has been demonstrated, however, that spherical amalgams are less sensitive to variations in condensation pressure than the amalgams containing nonspherical particles.[175] In addition, the spherical materials generally have a shorter working time and demonstrate a faster set than the admixtures.

Trituration

The trituration process includes the combining or mixing of liquid mercury with dry amalgam alloy powder. Electric amalgam mixers (also called *amalgamators* and *amalgam triturators*) are used for the trituration process (Fig 12-53). The objective is to remove the oxide coating and wet each particle of alloy with mercury.[8] This begins the reaction that will produce a solid mass. Although amalgam alloy pellets and bottled mercury are still available separately, the use of precapsulated amalgam alloy (that is, a weighed, standardized amount of amalgam powder and mercury sealed into a capsule) is strongly recommended. The precapsulated products are not only ready for trituration, but they also provide more consistent mixes of amalgam and virtually eliminate the possibility of mercury spills in the dental office.



Fig 12-53*(a)* Pro-Mix amalgamator (Dentsply/ Caulk). *(b)* Optimix amalgamator (Kerr/Sybron).

The duration and speed of trituration should be just enough to coat all alloy particles with mercury, produce the amalgam matrix, and provide a plastic mix. Excessive trituration should be avoided, because it generates heat that will cause the amalgam to set prematurely after trituration, and this will prevent adequate condensation and adaptation to the walls of the preparation, resulting in a weakened product. A mix of amalgam that is too plastic due to excess mercury or, as is more

frequently the case, is not plastic enough must be discarded. A good mix of amalgam is plastic enough to condense well. If the mix is too hard, brittle, or hot, reduction of the mixing time and/ or speed is indicated.

Condensation

Condensation is the process of compressing and directing the dental amalgam into the tooth preparation with amalgam-condensing instruments (called *condensers* or *pluggers*) until the preparation is completely filled, and then overfilled, with a dense mass of amalgam. Proper condensation promotes adaptation of the amalgam to the walls of the preparation, and it compacts the material, eliminating voids and reducing the amount of residual mercury in the restoration. Both voids and increased residual mercury have been associated with a weakened amalgam product, so effective condensation continues trituration[176] and increases the strength[175,176] and serviceability[12] of the restoration.

Adequate condensation technique requires that a significant amount of force be applied to the condenser.[177] The force should be 2 to 5 kg (5 to 10 lbs) for a condensable amalgam (admixture or conventional); the condensation force required for spherical amalgams will be considerably less,[175,176] because heavy forces with the condenser tend to push the spherical particles to the side and cause the condenser to "punch through" the amalgam mass. The size of the condenser nib (end) determines the amount of pressure actually transferred from the operator's hand to the amalgam mass; the larger the nib, the less force per unit area (pressure) is applied to the mass for a given force from the operator's hand. In other words, when a larger-faced condenser is used, the operator must exert more force on the condenser to deliver adequate condensation pressure. Larger condensers should be used for spherical amalgam, rather than for admixtures, to allow adequate force to be applied without displacement of the spherical amalgam to the side.

When amalgam bonding systems are used, amalgam must be condensed into the bonding resin on all walls of the preparation before filling of the preparation is begun. Adequate condensation force will cause a slight movement of the patient's mandible or head, and often this movement will need to be stabilized by the dentist or assistant. A secure finger rest will enable the operator to perform more controlled, forceful strokes, using arm as well as finger pressure. The condensers are held with the pen grasp or a modification. Many operators use a finger or thumb of the hand that is not holding the condenser to apply additional condensation force. It has been demonstrated that dentists tend to use less condensation pressure during

the later stages of amalgam placement; investigators[177] have emphasized the need for maintaining condensation pressures for both admixed and spherical amalgams throughout the condensation process.

After the preparation has been made ready to receive the amalgam, the amalgam alloy and mercury should be mixed (triturated) to give a plastic and moldable mass of amalgam. For most restorations, an amalgam carrier (see Fig 7-36) is valuable for delivering the amalgam to the preparation. For small restorations and for placing amalgam in proximal box preparations, care must be exercised to ensure that the increments of amalgam are small enough. If the increments are too large, the condensing force cannot be adequate to adapt the amalgam material at the deepest area of the increment. For very large complex amalgam preparations, the entire mix may be carried to the preparation, or the mix may be divided in half and carried to the preparation one half at a time with cotton forceps. No matter how the amalgam is carried, it should be spread in the preparation so that the increment is thin for optimum condensation. Each portion of amalgam carried to the preparation should result in an increment thickness of 1 mm or less to ensure maximum condensation effectiveness.

Condensers that fit into all areas of the preparation should be used. Flat-faced, round condensers are generally considered to allow maximum condensation pressure. Convex-ended condensers are also available, as are flat-ended condensers with diamond, rectangular, and triangular shapes. When spherical amalgams are used, the largest condenser that will fit into the area of the preparation where the amalgam is being condensed should be used. For all amalgams, a large condenser should be used for the overfilling of the preparation.

Amalgam must be condensed into the preparation as soon as trituration is completed. One increment of amalgam should not be allowed to set significantly before the next increment is added. Amalgam should be condensed both vertically and horizontally or laterally (toward the walls of the preparation). This will promote a close adaptation of the amalgam to the walls as well as to the floor of the preparation. Lateral condensation, whether or not an amalgam bonding system is being used, can be achieved in more than one way. One is to alter the direction of the face (end) of the condenser so that it is pushed toward the walls. Another method is to place the condenser into the preparation vertically, then to move it laterally toward the walls so that the side of the condenser condenses the amalgam against the walls (Fig 12-54a). Lateral condensation is especially important for spherical amalgams because, paradoxically, it is more difficult to adapt these materials to cavity walls.

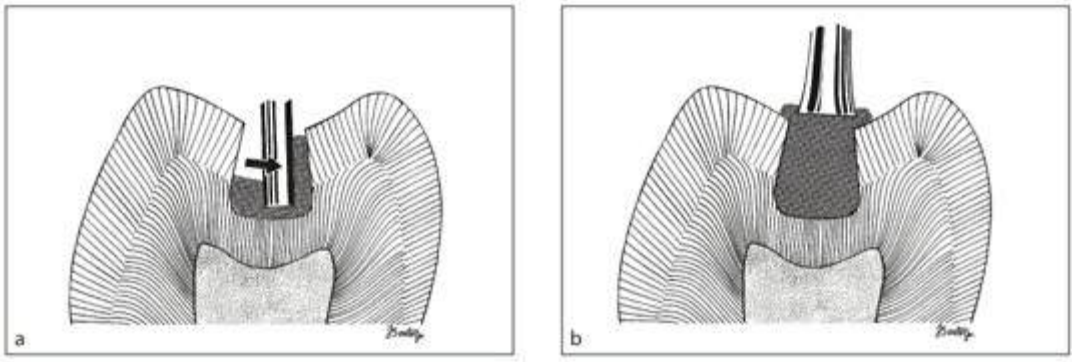


Fig 12-54*(a)* Lateral condensation toward all walls and toward the adjacent tooth in a Class 2 restoration will improve adaptation to the walls and ensure a contact area with the adjacent tooth. *(b)* Overfill should be condensed with a large condenser.

When amalgam is condensed, mercury tends to be brought to the surface, creating a mercury-rich amalgam on the surface. To reduce the amount of mercury left in the restoration (residual mercury), the preparation is overfilled (Fig 12-54b), and the mercury-rich excess is carved off. The lower the residual mercury in the carved restoration, the greater its strength[2] and the better the expected longevity of the restoration.

Condensation when amalgam bonding resins are used

As with the use of the matrix, there are some additional considerations and slightly altered techniques when amalgam bonding systems are used. Because the polymerization of amalgam bonding resins is chemically initiated, the amalgam must be ready to place when the two parts of the bonding resin are mixed to initiate polymerization. Although all walls of the cavity preparation should be coated, caution should be exercised to minimize the amount of bonding resin placed on the walls. One study demonstrated that a thin coat of amalgam bonding resin provided attachment of the restoration to tooth structure that was as strong as that provided by a thick coat, but without the problems caused by a thick coat.[178] One problem resulting from excess bonding resin is the reduction of the amalgam strength by incorporation of large amounts of resin into the bulk of the amalgam.[179] Another is the increased probability of transferring the bonding resin to the metal matrix during condensation, as described earlier. Another potential problem is the creation of voids in the proximal amalgam due to excess resin being pulled from the restoration during matrix removal.

After the bonding resin has been applied to the walls of the preparation, the amalgam is placed in the preparation and condensed against all walls within 1

minute. After amalgam has been condensed into the resin on all walls, it should be added in increments as described for nonbonded amalgam restorations.

Precarve burnishing

After it is condensed with amalgam condensers, the amalgam may be further condensed and shaping of occlusal anatomy begun with a large burnisher, such as an ovoid (football) bur-nisher (see Fig 12-62h). This is called *precarve burnishing*, and it should take place immediately after the completion of condensation. The burnisher should be used with heavy strokes, made in the mesiodistal and faciolingual directions, that pinch much of the amalgam off as the burnisher contacts the cusp inclines and, in some places, the margins of the preparation. It has been shown that precarve burnishing produces denser amalgam at the margins of restorations.[180] In addition to aiding condensation, precarve burnishing is the first step in shaping the occlusal surface of the restoration.

Carving

Amalgam can be carved with any bladed dental instrument that has a sharp edge. Numerous carvers are available, and each has its own merit. Recommended amalgam carvers that satisfy most amalgam carving needs include a small cleoid-discoid carver, a Walls no. 3 (or Tanner no. 5) carver, a Hollenback no. ½ carver, an interproximal carver, and a no. 14L sickle-shaped carver (see Fig 7-39). In addition, some cutting instruments, such as a small spoon excavator and hoe, make excellent amalgam carvers, especially for carving occlusal anatomy in large restorations. The carving instruments selected should allow the operator to create contours and occlusion that reproduce, or occasionally make improvements to, the missing tooth structure.

Carving can begin immediately after condensation and precarve burnishing. Before the setting of the amalgam is very advanced, amalgam carves very easily, but it is also easy to miscarve or overcarve, so care must be taken. As the setting of the amalgam advances, it does not carve as easily, but it remains carvable with sharp carvers for a long time. In fact, amalgam that has been in the mouth for many years can still be carved with sharp carvers. The need for sharp carvers cannot be overemphasized, and it is advisable to have a sterilized sharpening stone available during placement of a large amalgam restoration. Amalgam seems to cause rapid dulling of carvers, possibly because of the effect of the mercury in penetrating and imparting brittleness to the steel.[181]

Most occlusal carving is performed with pulling strokes, but the pushing stroke can also be advantageous in developing occlusal anatomy. Smaller occlusal and Class 2 restorations should be carved with the enamel tooth surface as a guide (Fig 12-55). The carver should rest on the enamel adjacent to the preparation and be pulled in a direction parallel to the margin of the preparation. When a stroke that is perpendicular to the margin of the preparation is needed, the carver should be pulled from enamel to amalgam. If it is pulled from amalgam to enamel, it will be more likely to carve the surface of the amalgam to a level that is below the surface of the enamel. It is desirable that the two surfaces be even (at the same level) so that there is no "step down" from the enamel to the amalgam.

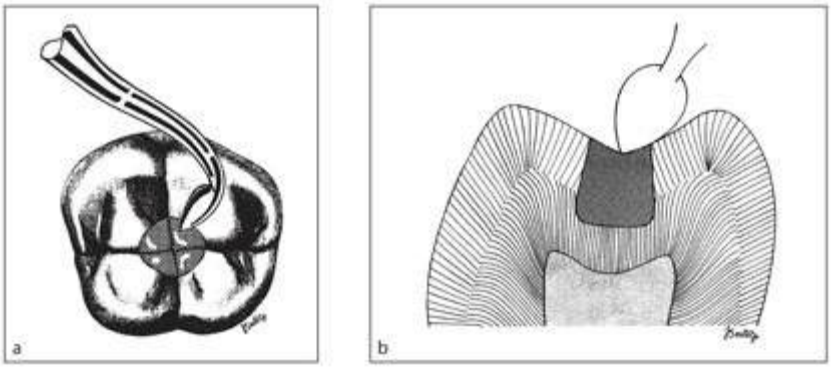


Fig 12-55 The enamel margin is used as a guide for carving smaller restorations. *(a)* Cleoid carver viewed from the occlusal aspect. *(b)* Cleoid carver viewed in cross section.

It is good to register a mental picture of the outline of the preparation before the amalgam is placed so that the outline can be visualized after carving. Amalgam preparations should have enamel margins that are not jagged or rough; if the margins of a carved restoration appear ragged, it will be because of thin amalgam flash that extends outside of the preparation onto the adjacent enamel surface. This flash is more difficult to remove when amalgam bonding resins are used. A sharp carver is even more necessary for effective removal of this flash.

Amalgam should not be overcarved such that groove anatomy is deep, leaving thin fins of amalgam adjacent to the preparation margins. The operator should try to develop margins that will leave a 75- to 90-degree angle at the margin of occlusal amalgam. Acute angles (fins) of amalgam at the margins on an occlusal surface are subject to fracture during function.

For Class 2 restorations, while the matrix is in place, the marginal ridge should be carved very nearly to the height of the adjacent marginal ridge (see Fig 12-62i). Development of the occlusal embrasure of the marginal ridge is begun with the tip of an explorer angled at approximately a 45-degree angle to the long axis of the tooth

and touching the matrix band (Fig 12-56; see also Fig 12-62j). The explorer tip should be moved from the facial enamel, past the margin of the box, to the center of the marginal ridge and then from the lingual enamel, past the margin of the box, to the center. The explorer should not be moved from the amalgam toward the margin, because this movement could easily result in overcarving, leaving the marginal ridge with a deficient contour.

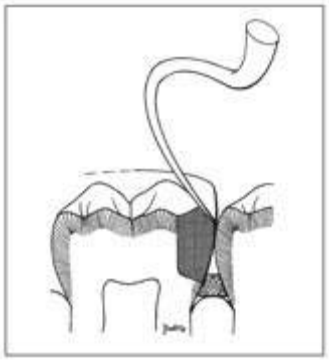


Fig 12-56 The tip of the no. 23 explorer is used at a 45-degree angle to the matrix to begin shaping the occlusal embrasure.

Most carving will be accomplished while the rubber dam is in place. For Class 2 restorations, after the matrix is removed, amalgam flash on proximal surfaces should be removed and the proximal contours should be refined. A thin carver, such as the interproximal carver, is useful for both removing flash and refining proximal contours (Fig 12-57; see also Figs 12-62w to 12-62y). Because the proximal contour is so crucial for periodontal health,[182] removal of the matrix is strongly advocated early in the carving process. While the amalgam is still soft, proper contours, as well as removal of excess amalgam from proximal margins, can readily be achieved. As previously stated, if a bonding system has been used, it is important to minimize the amount of bonding resin at proximal margins to prevent marginal voids that can be produced if the resin at margins is pulled out during carving. A very sharp carver will slice this incompletely set resin instead of catching and dislodging it.

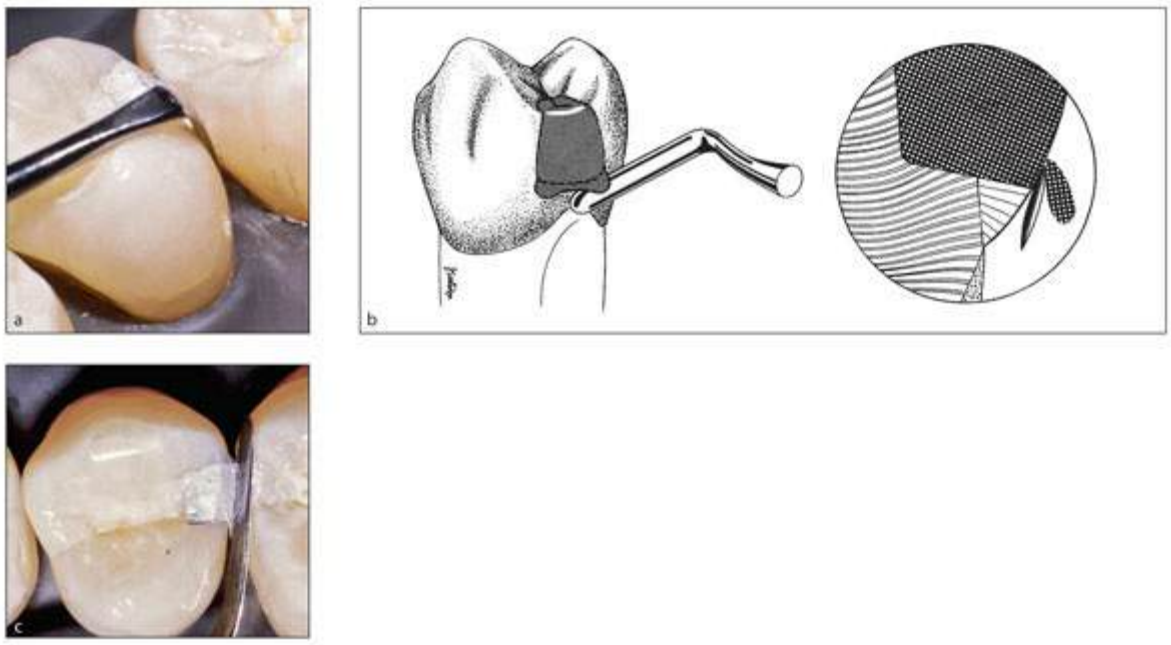


Fig 12-57*(a to c)* An interproximal carver is used to remove flash and to contour and burnish the amalgam in interproximal areas.

Adjusting the occlusion

When the carving appears to be correct, the dam is removed, and the occlusion is checked. This is accomplished with articulating ribbon (Fig 12-58), which marks the points of contact when the mandibular and maxillary teeth are brought together. It is wise not to ask the patient to close, because, if the amalgam has not been carved adequately, it will be "high" in the occlusion so that it contacts first, prior to any other tooth contact. The masseter muscles are very strong, and when the proprioceptive innervation relates to the patient's brain that there is something between the maxillary and mandibular teeth, it is reflex action for the patient to attempt to masticate it. In the case of a high amalgam restoration, disaster can result; the amalgam will usually be fractured, and the operator will have to remove amalgam and begin again.

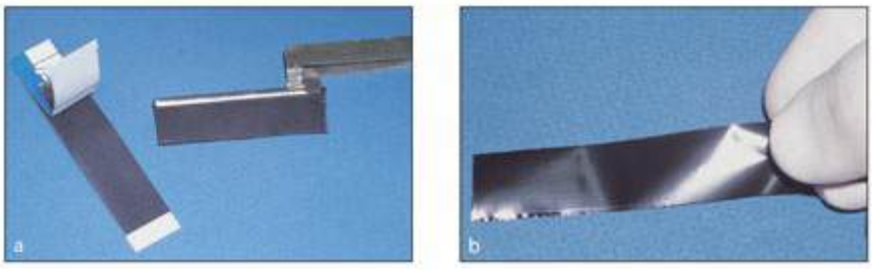


Fig 12-58*(a)* For initial gross adjustment, a piece of articulating paper with a thickness of 20 µm (0.0008 inch) or more is useful. When articulating paper forceps are used, the total length of the piece of articulating tape or paper should be supported by the forceps. *(b)* For refining occlusion,

especially in complex amalgam restorations, an articulating tape with a thickness of 15 μm (0.0006 inch) or less is advantageous.

It is therefore best for the operator to perform the tapping of the teeth by grasping the patient's chin, having the patient close to very near contact, and then, by hand, manipulating the mandible so that mandibular and maxillary teeth are tapped together in maximum intercuspation position (MIP). The dentist's arm, no matter how strong, will be unable to impart nearly as much force in mandibular closure as the masseter muscles are capable of achieving. An alternative to this tapping by the dentist is to instruct the patient to "very, very gently, tap the back teeth together."

The amalgam must be carved until contacts on the restoration occur simultaneously with other occlusal MIP contacts on that tooth and adjacent teeth. These can be seen as marks made by articulating ribbon, but they should also be felt by the dentist with 0.0005-inch (12-μm) thick shimstock (Artus) (Fig 12-59). To do this, the patient should be instructed to close in MIP ("bite the back teeth together") while shimstock is in place on the tooth being restored. With the teeth in maximum intercuspation (MI), the shimstock should be held securely in place. The same test should be performed with the shimstock on adjacent teeth, and it should again be held securely (assuming that those teeth held shimstock prior to the restorative procedure). If the adjacent teeth do not hold the shimstock, the newly placed restoration is probably in hyperocclusion and needs additional carving.

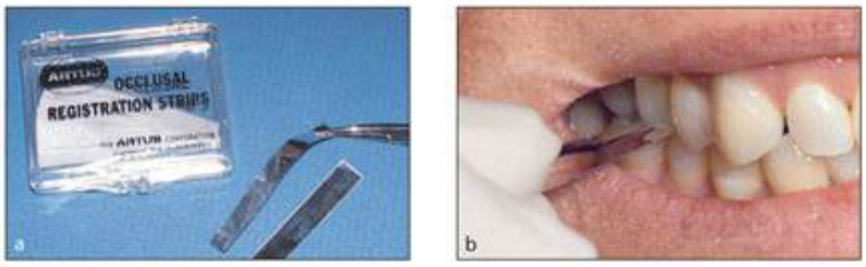


Fig 12-59 *(a)* Silver-colored shimstock (0.0005-inch-thick Mylar) is supplied in books with paper separators between the pieces. It may be held in the fingers or with a hemostat (as shown at right). *(b)* Shimstock is used to "feel" contacts between maxillary and mandibular teeth.

When the restoration occlusion is correct in MI, it must be checked to ensure that no interferences are caused by the restoration in excursive movements (laterotrusive, mediotru-sive [right or left], or protrusive movements) of the mandible. This may also be evaluated with the use of the shimstock. Two colors of articulating ribbon (preferably ribbons that do not easily cause smudge marks) are used, one color for latero-/ mediotrusive and protrusive excursions and the other to mark MI occlusal contacts. To eliminate excursive contacts in the amalgam, the amalgam marked with

the color used to register the excursions should be carved, and the amalgam marked with the MI marks should be preserved. For complex amalgam restorations, it should be ensured that the restoration does not cause interference in the slide between centric occlusion and MI.

Postcarve burnishing

Postcarve burnishing is the light rubbing of the surface of a carved amalgam restoration with a burnisher, such as the PKT3 (P. K. Thomas). Heavy forces should not be used, and postcarve burnishing should be avoided near the margins of restorations of fast-setting amalgam.[183] The purpose of postcarve burnishing is to smooth the surface of the restoration.

After completion of carving and postcarve burnishing, if the carving time was short and the amalgam is still fairly soft, the surface may be wiped over with a dry or water-damp cotton ball or cotton roll to provide additional smoothing. If the set of the amalgam is advanced so that the cotton will not smooth the surface, a rubber prophylaxis cup with damp flour of pumice or prophylaxis paste will smooth the amalgam (see the section on finishing and polishing). If the cup is used, it should be rotating at a very low speed and should be kept moving at all times; if the cup is allowed to rotate in one place, it will groove the recently carved amalgam.

Placement of amalgam in complex preparations

Several special considerations for placing complex amalgam restorations, such as the possible need for reinforcement of matrices with compound, have already been discussed. Following are some other considerations that may help the operator to successfully place complex amalgam restorations:

- Visualize the finished product, and shape the matrix to allow for that product.
- In addition to visualizing the height of the cusp tips before making the tooth preparation and building cusps back to that height, make a mental picture of facial and lingual contours before cutting away natural tooth structure so that you can reproduce the natural contours in amalgam as closely as possible. Note the contours by viewing down the line of teeth (in profile) from the facial and/or lingual aspect. Make sure that the final contours harmonize with the contours of other teeth in the quadrant (Fig 12-60; see also Fig 12-62ee).

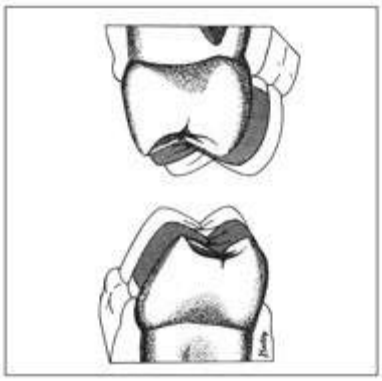


Fig 12-60 Facial and lingual contours are noted by looking down the line of teeth in a quadrant. The contour of the restoration must harmonize with natural tooth contours in the quadrant.

- Place larger increments of amalgam (eg, the entire two-spill [600-mg] mix), when replacing the entire occlusal surface of a molar or a half mix for less extensive restorations.
- Consider the use of carvers that contribute to proper contours, such as the no. 14L carver, which simplifies the carving of convex axial contours (Fig 12-61; see also Fig 12-62q).

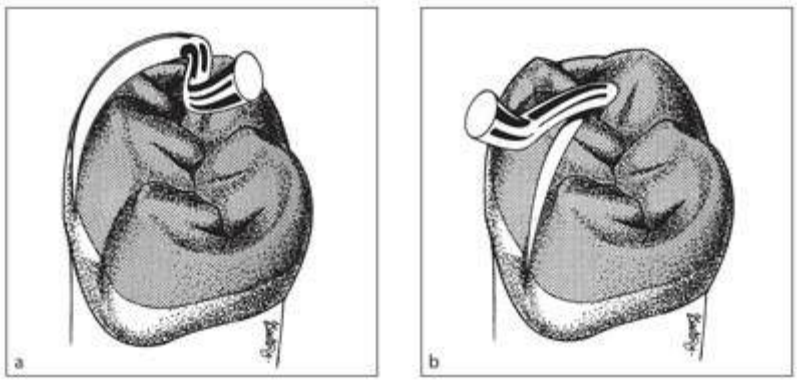


Fig 12-61*(a)* A no. 14L sickle-shaped carver will carve a very convex surface if it approaches the surface to be carved at just less than a 90-degree angle. *(b)* The same instrument will carve a surface with less convexity if it is rotated so that it approaches the surface at an angle of much less than 90 degrees.

- Because carving of large amalgam restorations involves the carving of more surface area, consider sharpening the carvers during the procedure to allow for more efficient carving.
- Smooth the carved amalgam with a water slurry of flour of pumice or with a prophylaxis paste.

A series showing the insertion of a complex amalgam restoration, beginning with matrix application and ending just before rubber dam removal, is shown in Fig 12-

62. Several amalgam preparations and restorations are shown in Figs 12-63 to 12-74.

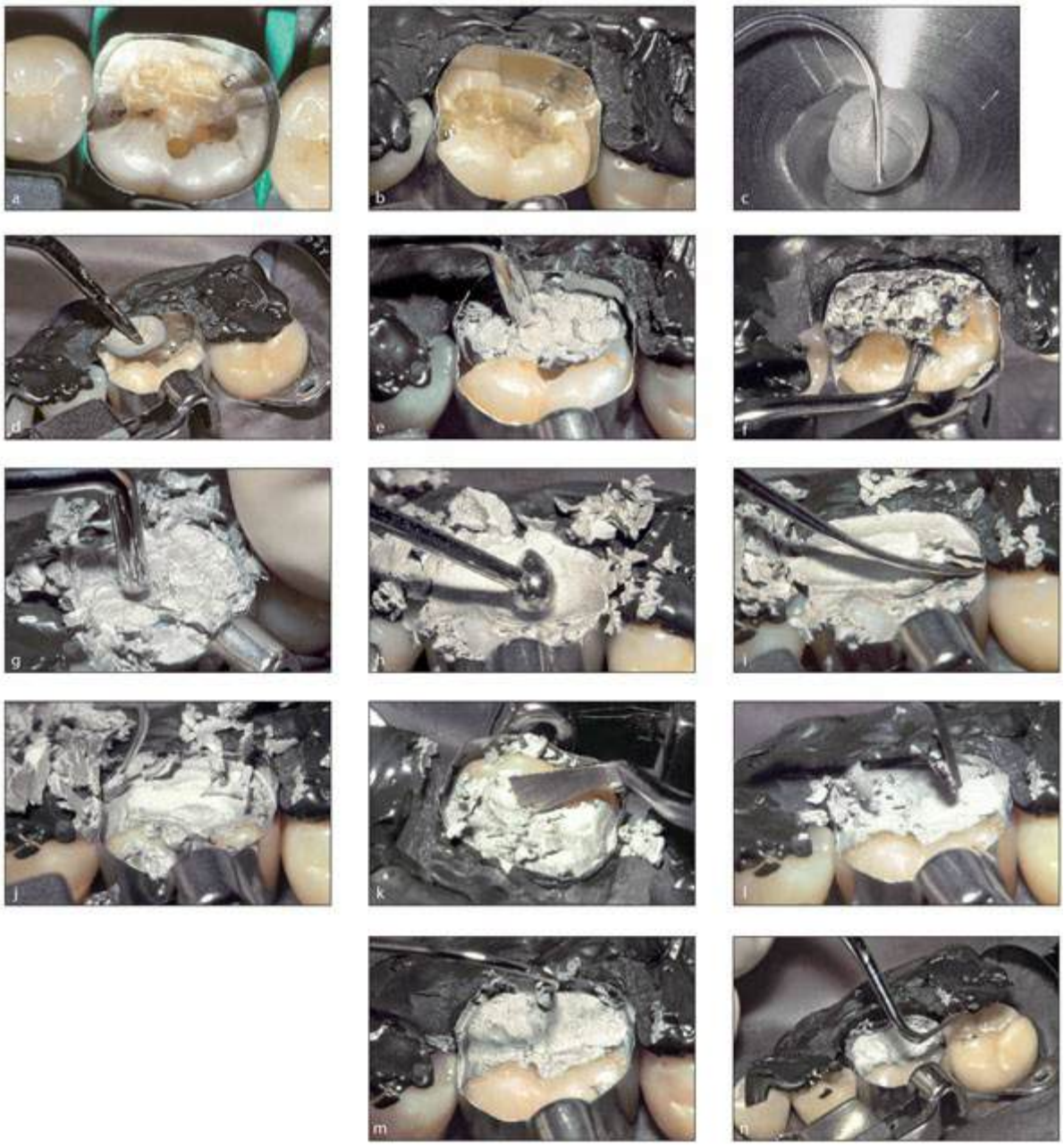


Fig 12-62 Condensation and carving of a complex amalgam restoration. *(a)* A Tofflemire matrix is placed and shaped to provide the desired contours. *(b)* The matrix is stabilized with modeling compound. *(c)* A two-spill mix of amalgam is halved. *(d)* Half of the mix of amalgam is carried to the preparation. *(e)* Amalgam is spread over the entire preparation floor and condensed. *(f)* A small condenser is used to condense amalgam into the amalgapin channel. It should also be used to condense amalgam into internal line angles and at corners created by the matrix band at cavosurface margins. *(g)* Amalgam increments are added in 1-mm thicknesses until the preparation is overfilled. *(h)* Amalgam shaping is begun with a large ovoid burnisher used to pinch excess amalgam off against the enamel (precarve burnishing). *(i)* Marginal ridge shaping is begun by reducing amalgam in the area to the approximate desired height of the marginal ridge. *(j)* The occlusal embrasure is formed with an explorer tip. *(k and l)* A chisel-shaped carver (Walls no. 3)

is used to begin shaping cusps and grooves while the matrix is still in place. *(m)* An explorer tip is used to begin shaping the lingual contour inside the matrix. *(n)* The compound and matrix are removed.



Fig 12-62*(cont) (o to q)* A sickle-shaped carver (no. 14L) is used to remove gingival flash, shape proximal surfaces, and shape lingual contour. *(r and s)* The marginal ridge is adjusted to height by resting the carver (Walls no. 3) on the adjacent marginal ridges during carving. *(t to v)* The occlusal anatomy is refined with a hoe. *(w and x)* The proximal contours and contact position are refined with a very thin interproximal carver. *(y)* The occlusal embrasure is refined with an interproximal carver, which is rested on the adjacent enamel to guide the marginal ridge contour. *(z)* The surface is smoothed with a medium-grit prophylaxis paste in a rubber cup.

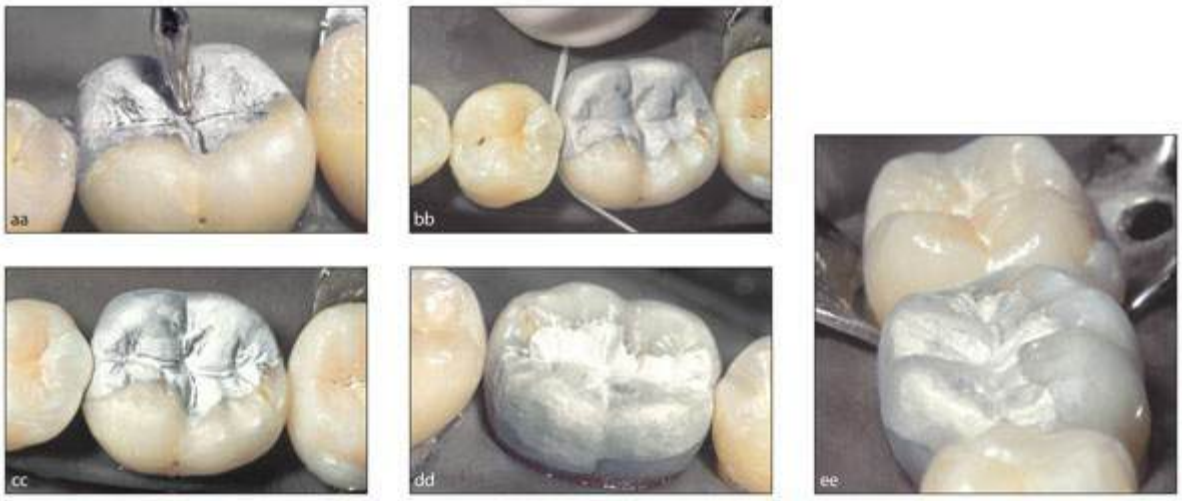


Fig 12-62*(cont) (aa)* The bases of the grooves are smoothed with a burnisher (PKT3). *(bb)* The proximal contours are "felt" with floss to ensure smoothness and to clear any amalgam carvings from the contact. *(cc to ee)* Carving is completed. Note that the lingual contour harmonizes with the contours of the adjacent teeth. The rubber dam is then removed, the occlusion is refined, and the surface is resmoothed with pumice or a prophylaxis paste.



Fig 12-63*(a)* A small area of carious dentin and overlying unsupported enamel has been removed to complete the occlusal amalgam preparation. *(b)* The preparation is filled with amalgam. *(c)* The remaining occlusal fissures were opened with a no. ¼ bur to a depth of 0.5 mm, etched, and sealed with resin fissure sealant.

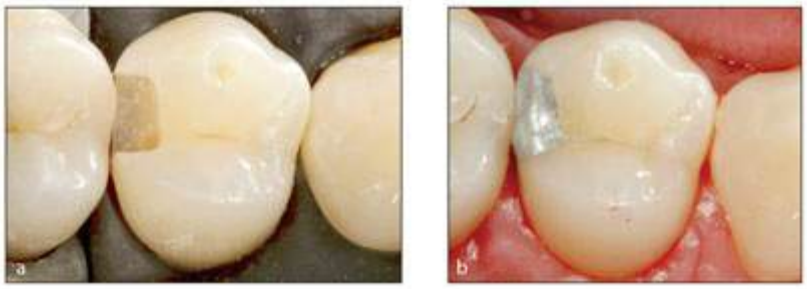


Fig 12-64*(a)* Disto-occlusal slot preparation with retention by bonding. *(b)* Slot restoration. There was no treatment of occlusal fissures.

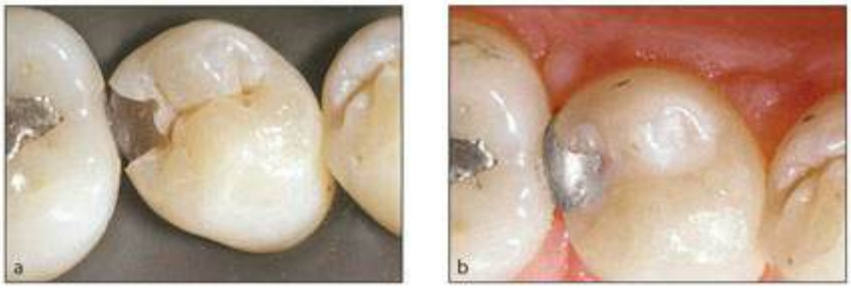


Fig 12-65*(a)* Disto-occlusal slot preparation with retention grooves. *(b)* Slot restoration. The occlusal fissure was sealed.

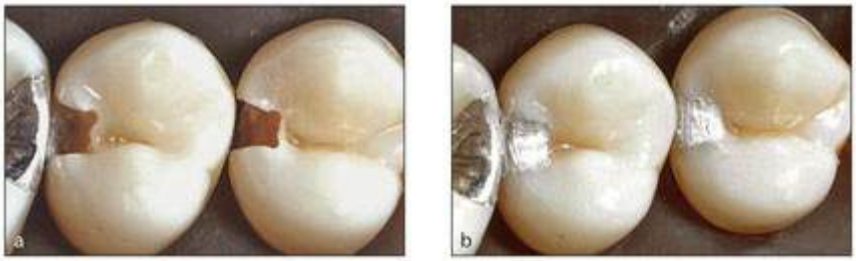


Fig 12-66*(a)* Disto-occlusal slot preparations. *(b)* Slot restorations.



Fig 12-67*(a)* Small occlusal and disto-occlusal slot amalgam preparations. *(b)* Restorations with sealed occlusal fissures. *(c)* Restorations at 2 years.

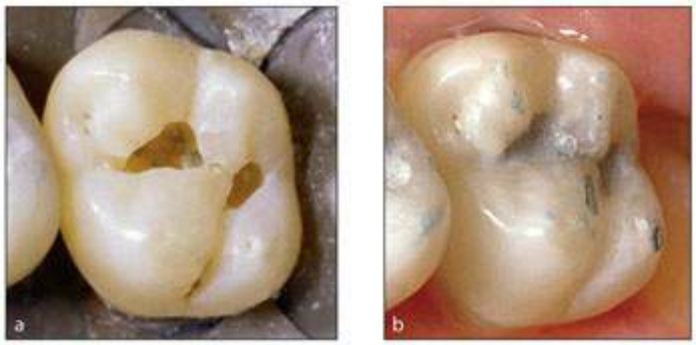


Fig 12-68*(a)* Occlusal preparations. *(b)* Restorations with remaining occlusal and lingual fissures sealed.

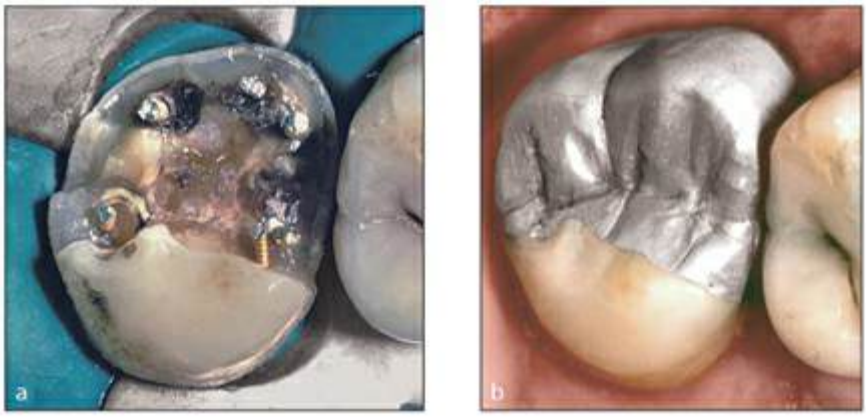


Fig 12-69*(a)* Complex amalgam preparation with vertical and horizontal pins in the molar and a disto-occlusal slot preparation in the premolar. *(b)* Restorations with sealed occlusal fissures.



Fig 12-70*(a)* Old resin composite restoration with recurrent caries. *(b)* Complex amalgam preparation with six vertical pins. *(c)* Complex amalgam restoration.

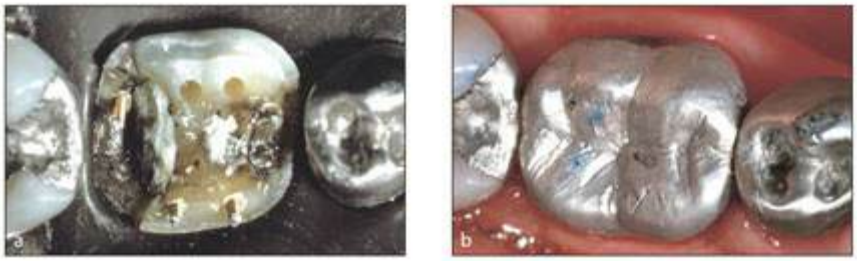


Fig 12-71*(a)* Complex amalgam preparation with pins, boxes, and amalgapins. *(b)* Complex amalgam restoration.



Fig 12-72*(a)* Complex amalgam preparation with boxes, amalgapins, and a shelf. *(b)* Complex amalgam restoration. *(c)* Restoration at 3 years.

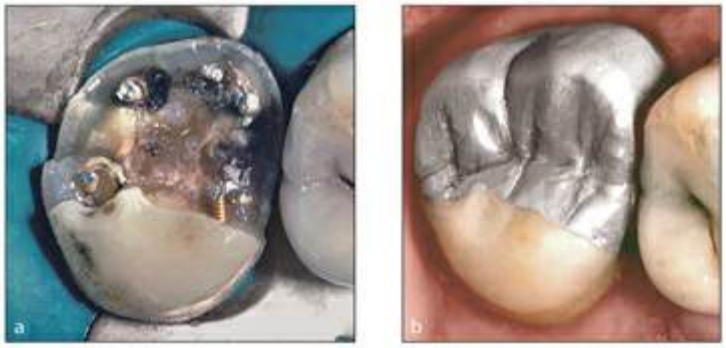


Fig 12-73 *(a)* Complex amalgam preparation with horizontal and vertical pins. *(b)* Complex amalgam restoration.



Fig 12-74 Complex amalgam restoration covering all cusps, with no mechanical retention form prepared; this restoration was bonded with a resin bonding system (Amalgambond Plus with HPA powder): *(a)* preparation; *(b)* completed restoration immediately after placement; *(c)* restoration at 6-year recall.

Finishing and Polishing Amalgam Restorations

Finishing of an amalgam restoration includes evaluating the restoration for problems and correcting them, ensuring that the margins of the amalgam restoration and the cavity-preparation walls are even and that the contours and occlusion are correct, and smoothing the restoration. *Polishing* is defined as smoothing the surface to a point of high gloss or luster. It has been demonstrated that polishing a high-copper amalgam restoration does not enhance its clinical performance,[184] but finishing is an important part of restoration placement. Finishing is usually accomplished at the placement appointment, but it may be refined at succeeding appointments.

Despite the lack of evidence that longevity is increased or performance improved when an amalgam restoration is polished, a high luster is often more comfortable to the patient's tongue than an unpolished surface, so polishing is sometimes desirable. There are no contraindications to polishing a restoration, but care must be taken not to create excessive heat during the polishing procedure. Excessive heat generation may be injurious to the pulp of a vital tooth.

After placement of a restoration, its surface should be rubbed with a burnisher or

with cotton until it is smooth. For amalgam with a more advanced set, a rubber cup with wet pumice or a prophylaxis paste may be used to smooth the restoration. Polishing of an amalgam restoration should be accomplished at a succeeding appointment, or at least some time after placement of the restoration. If an amalgam is adequately smoothed immediately after placement, imparting a high luster is usually a very simple and quick procedure.[185] If the restoration is not made smooth at placement, more time is required for polishing.

If, at the time of polishing, the restoration surface is not smooth, it should be smoothed. Gross smoothing of set amalgam can be accomplished with sharp amalgam carvers and finishing burs (Fig 12-75). For polishing convex surfaces (facial, lingual, and proximal), a series of progressively finer disks may be used (Fig 12-76). Alternatives for smoothing and polishing convex surfaces are the abrasive-impregnated rubber cups, using first the coarser cups and then the finer cups (Fig 12-77). Abrasive-impregnated rubber points are useful for smoothing and polishing concave surfaces, such as the occlusal surface. It is especially important that rubber polishers and abrasive disks are used with an abundance of air coolant and intermittent contact with the amalgam to prevent excessive generation of heat, which will bring liquid mercury to the surface of the amalgam, resulting in excessive volatilization of mercury and a weakened amalgam surface.

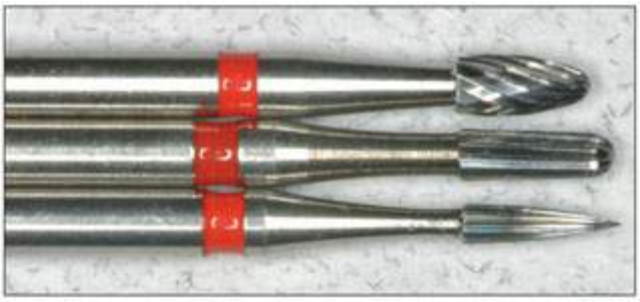


Fig 12-75 Finishing burs for a friction-grip handpiece: *(top to bottom)* no. 7404 (bud or egg shaped), no. 7803 (bullet shaped), no. 7901 (needle shaped).

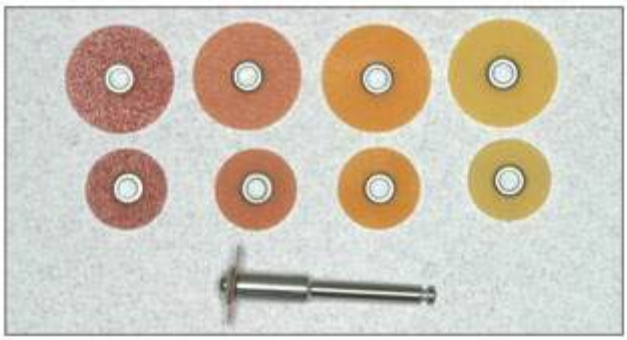


Fig 12-76 Abrasive disks, manufactured for polishing resin composite restorations, are also useful for polishing amalgam. The brown-yellow series of Sof-Lex disks and the pop-on mandrel (3M

ESPE) are shown. Also available is the thicker but more flexible black-blue series of Sof-Lex disks.



Fig 12-77 Brasseler abrasive-impregnated cups and points for polishing amalgam restorations: (*left to right*) coarsest (black), prepolish (brown), high shine (green), and super high shine (green with yellow band).

Although the disks and rubber polishers are more convenient, a less expensive, time-tested alternative method is the use of a prophylaxis cup, first with pumice in a water carrier as the "prepolishing" step, and then with tin oxide in a water or alcohol carrier for a high shine. One study[185] showed the pumice and tin oxide polishing procedure to be faster, but the investigator concluded that the impregnated points and cups are more desirable because they do not produce splatter.

A highly polished amalgam restoration is often more pleasing to the dentist than to the patient. A high polish can make a posterior amalgam restoration more noticeable, and this can be esthetically unpleasant to the patient. If this should occur, air abrasion with 50-µm aluminum oxide (Microetcher, Danville Engineering) or abrasion with pumice and a prophylaxis cup may be used to eliminate the high shine without making the restoration noticeably rough to the patient's tongue.

Repair of Amalgam Restorations

When an amalgam restoration has a defective area but the remainder of the restoration is adequate, a repair procedure may be the most appropriate treatment. For instance, if a cusp that was left in place adjacent to an amalgam restoration has fractured but the remaining amalgam restoration is serviceable, it might be appropriate to simply build a new cusp with amalgam. Or, if an amalgam fracture has occurred in the mesial box portion of a mesio-occlusodistal restoration but the remaining disto-occlusal portion involves a very gingivally deep distal margin, the

most conservative and simplest treatment might be to replace only the mesio-occlusal portion of the restoration.

Attachment of new amalgam to old can be achieved, but the attachment strength is only 30% to 60% of unrepaired amalgam.[82,186,187] Additional mechanical retention should be considered.

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